

BFS Traversal

Step 1: Start at A

Legend

Complete

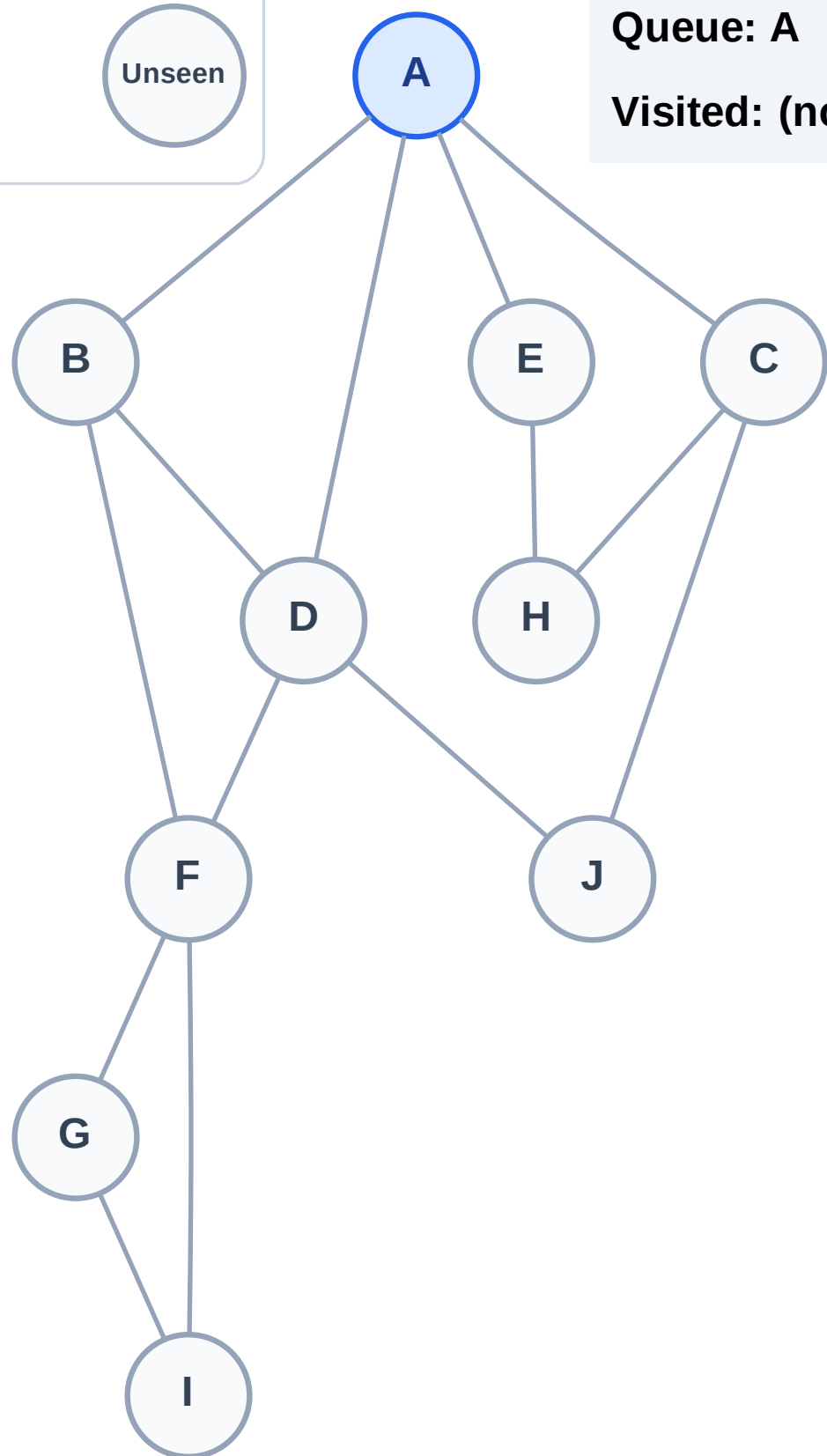
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

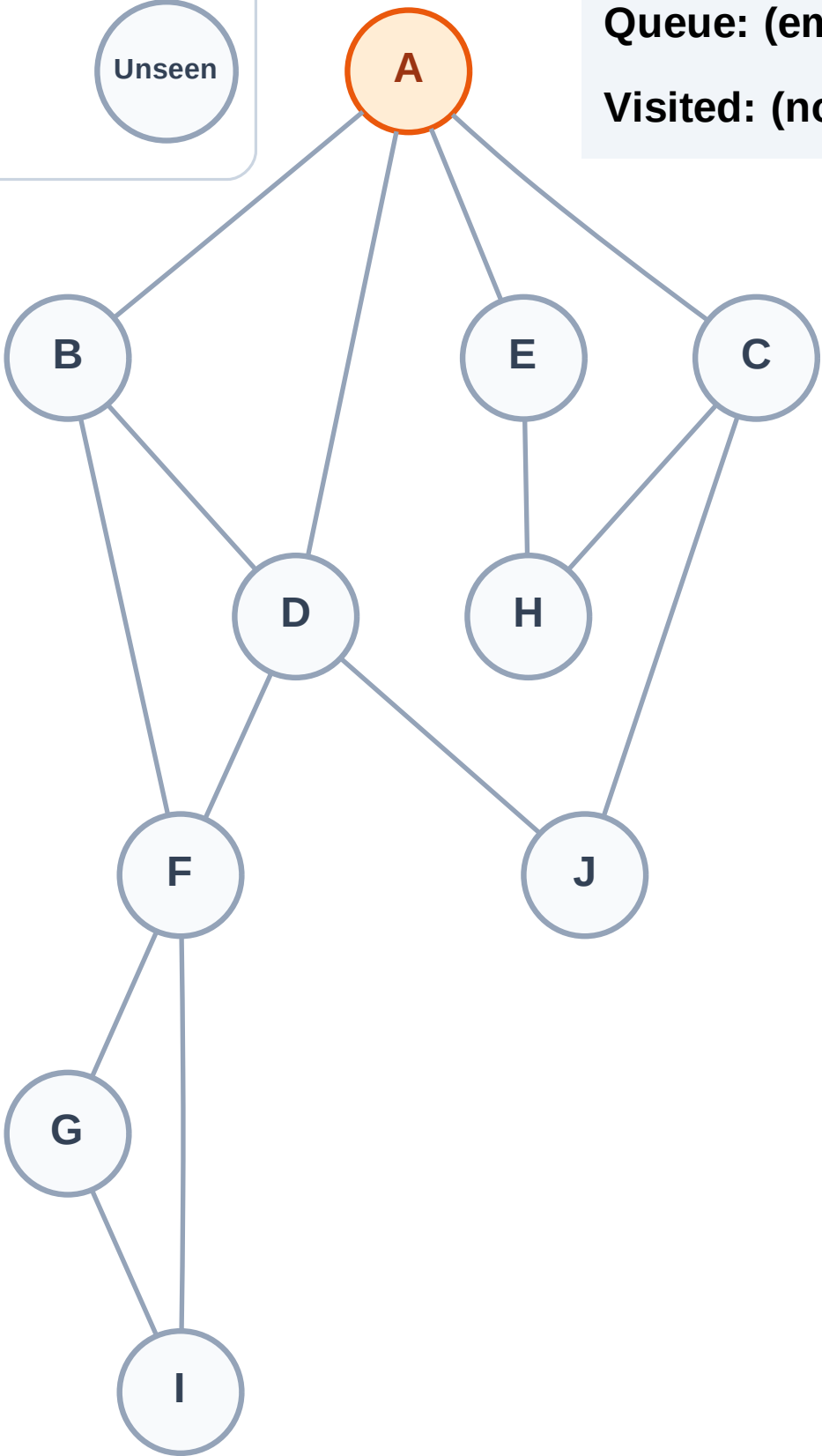
Step 2: Process node A

Legend



Queue: (empty)

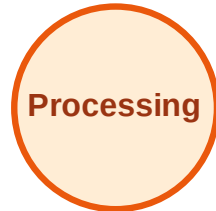
Visited: (none)



BFS Traversal

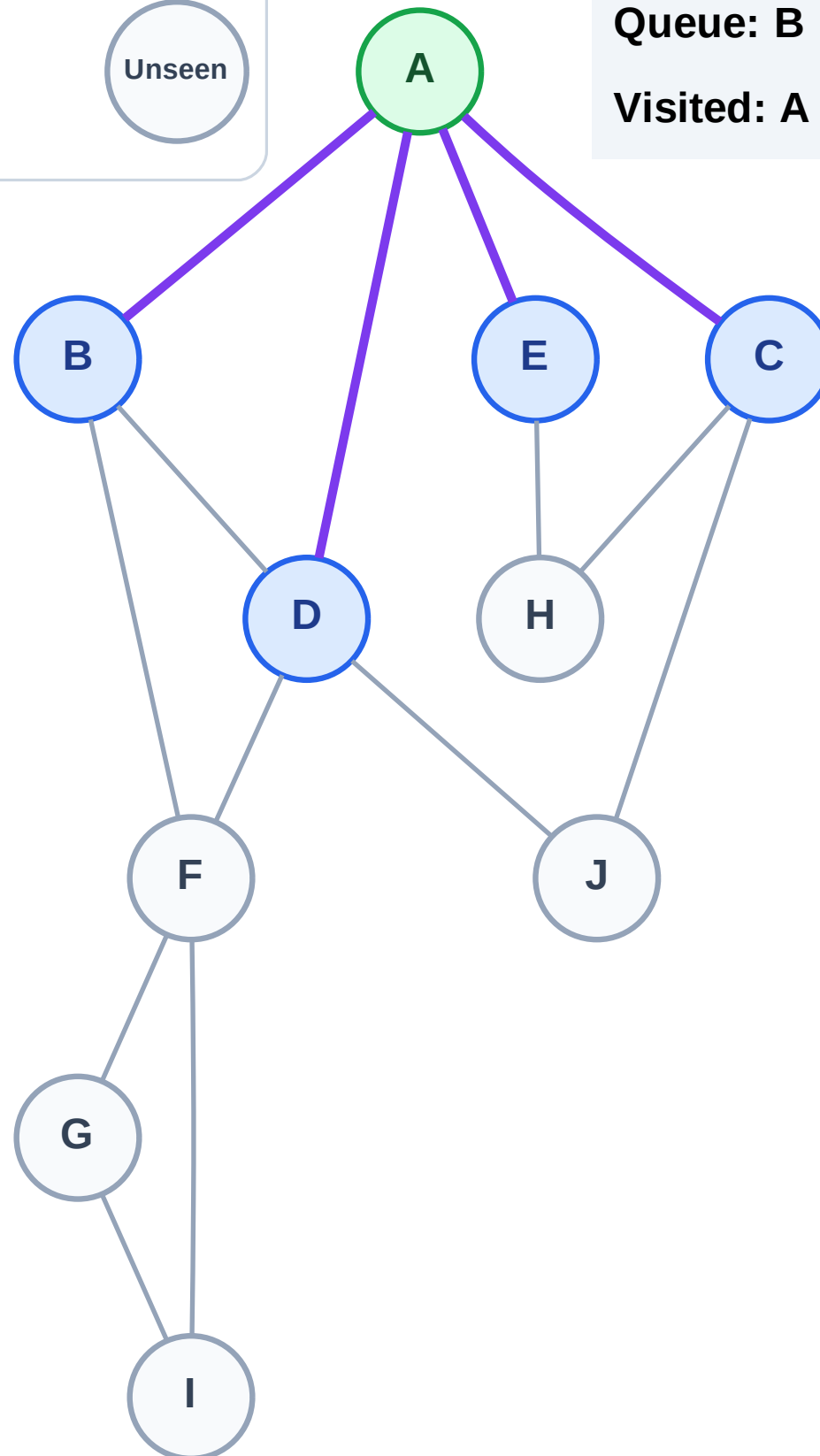
Step 3: Discover B, C, D, E from A

Legend



Queue: B → C → D → E

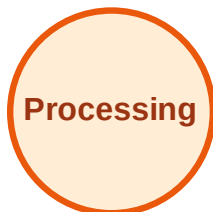
Visited: A



BFS Traversal

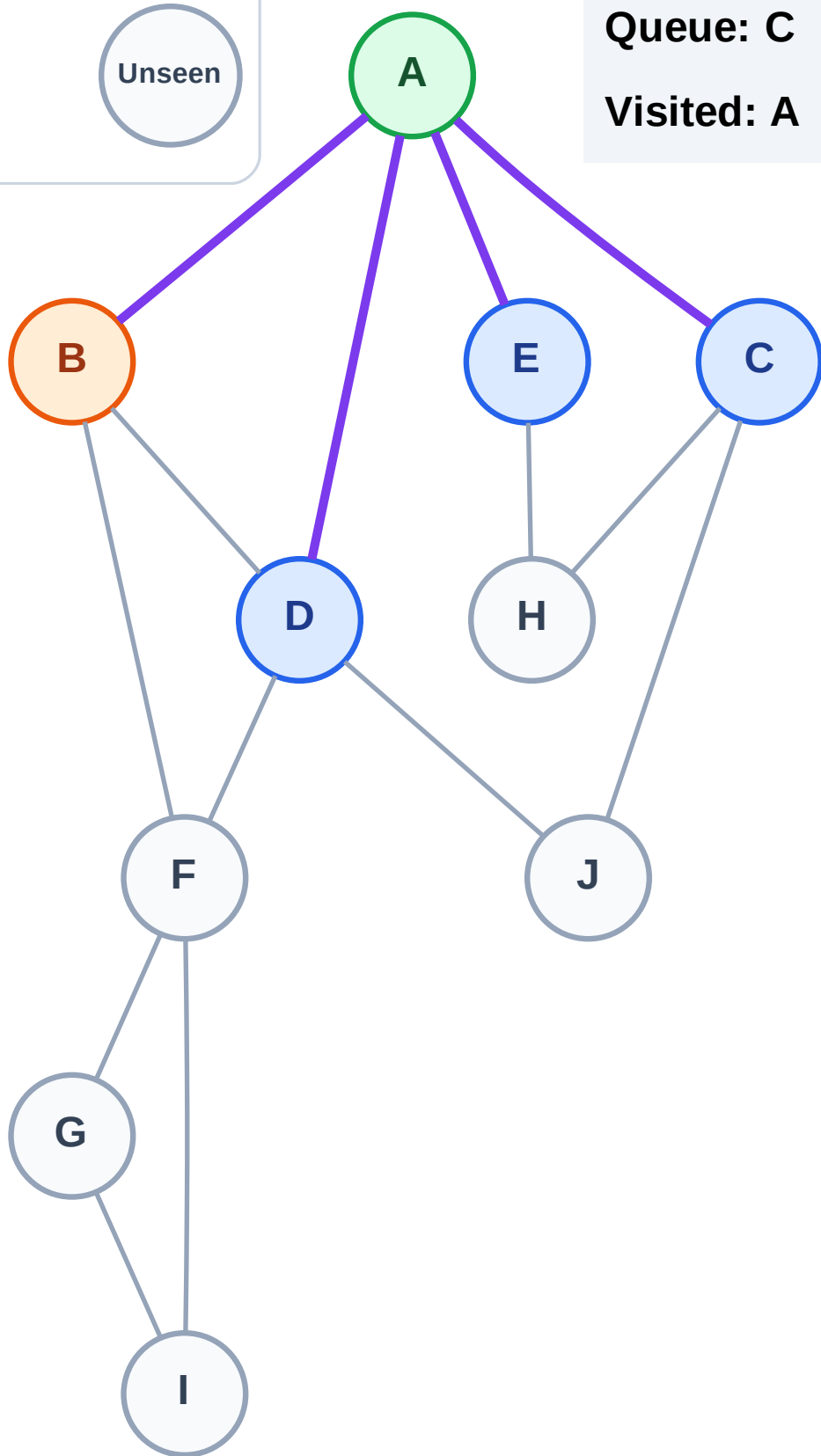
Step 4: Process node B

Legend



Queue: C → D → E

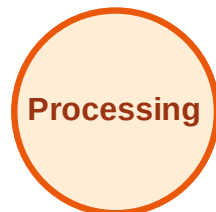
Visited: A



BFS Traversal

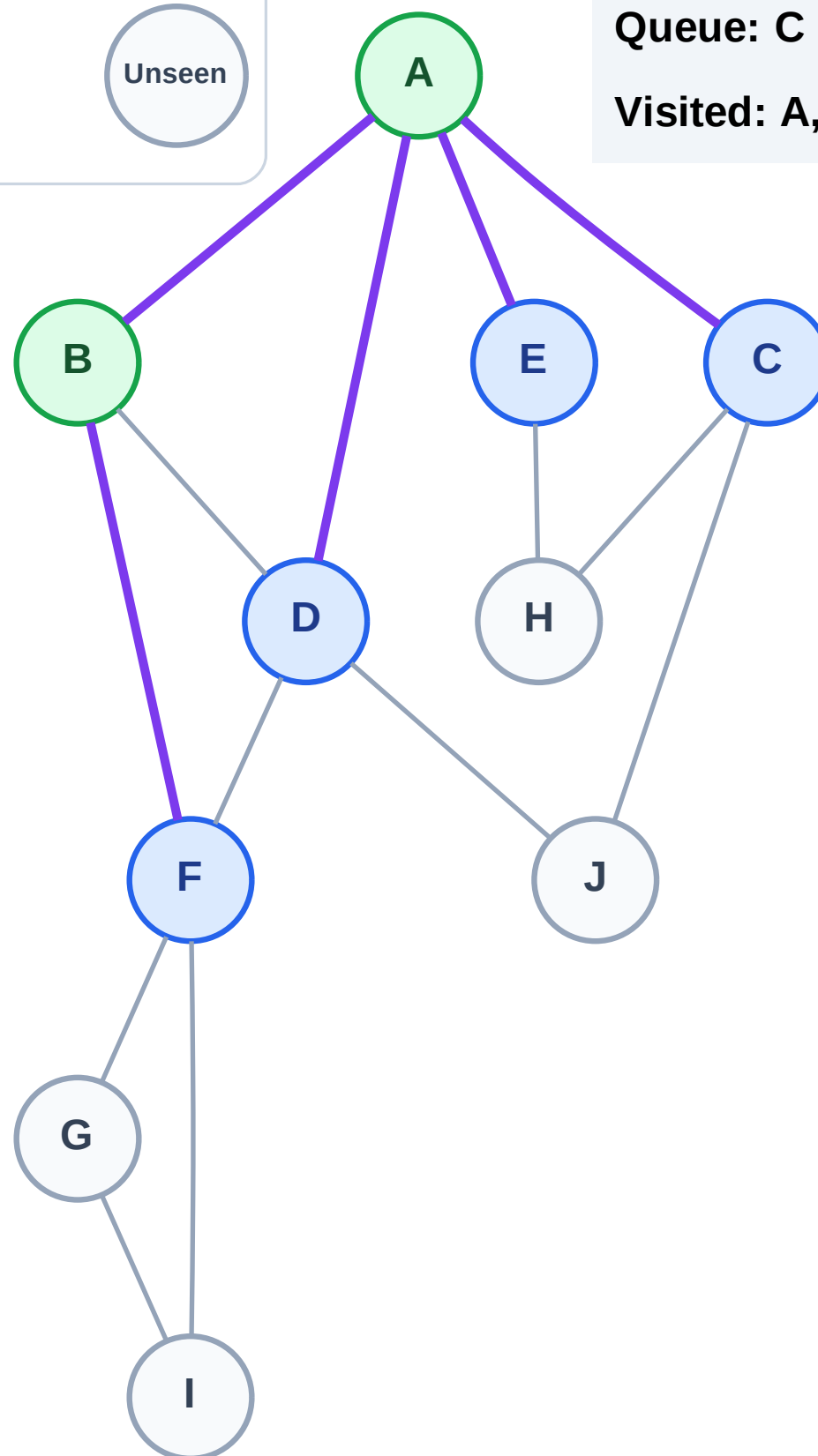
Step 5: Discover F from B

Legend



Queue: C → D → E → F

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

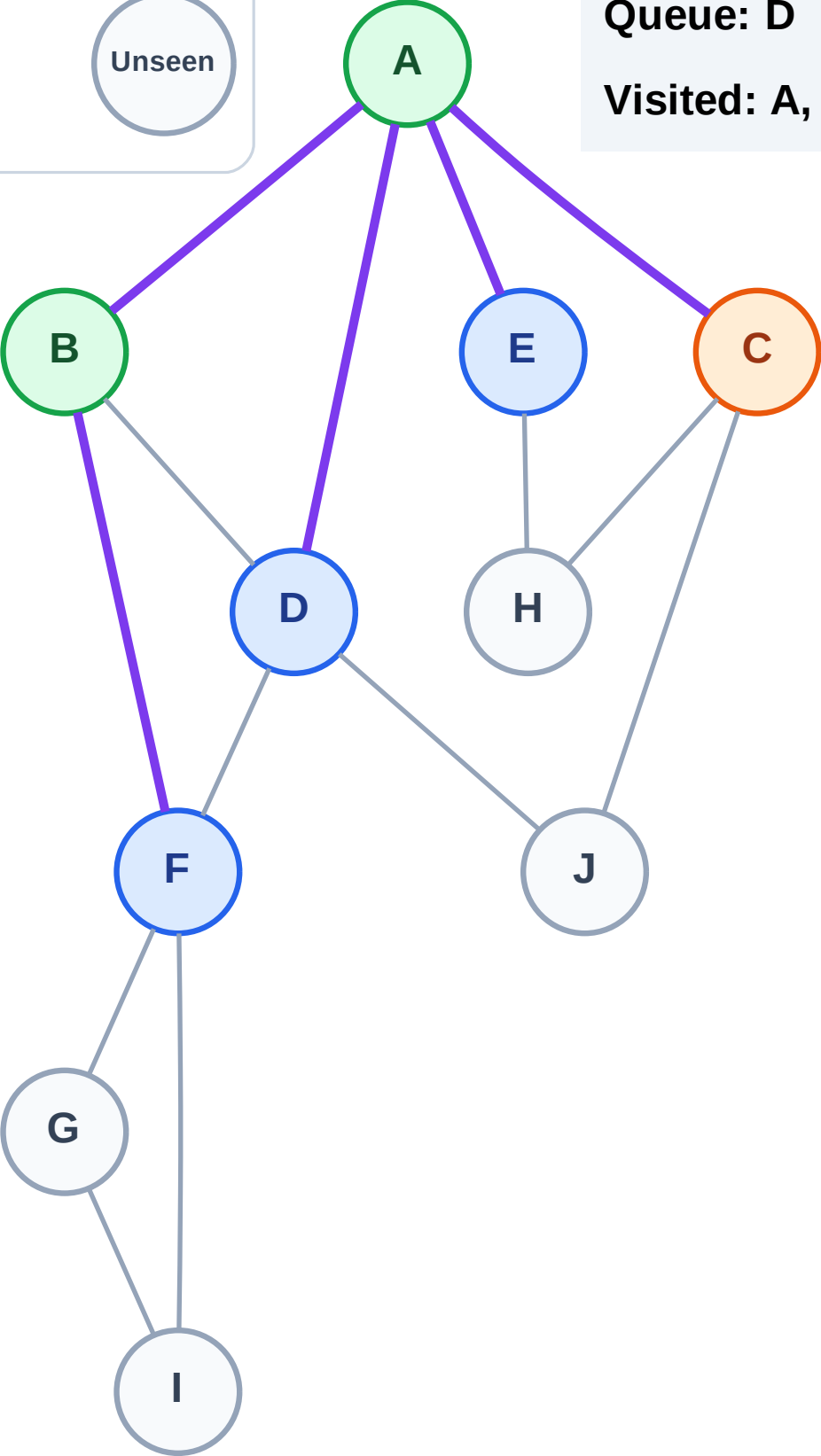
Complete

Processing

Discovered

Unseen

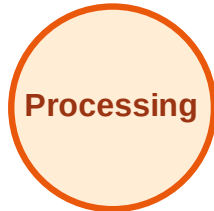
Queue: D → E → F
Visited: A, B



BFS Traversal

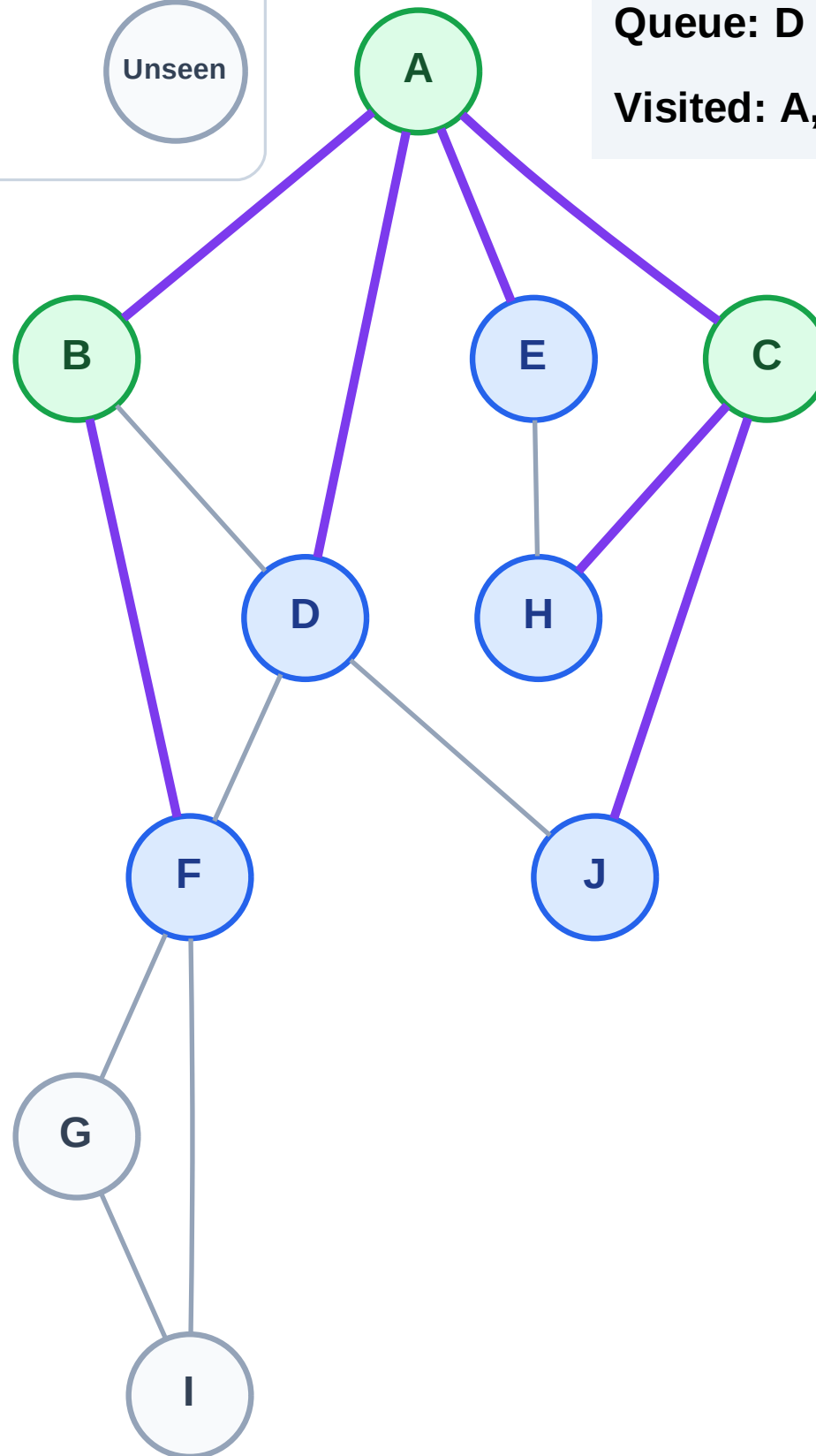
Step 7: Discover H, J from C

Legend



Queue: D → E → F → H → J

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

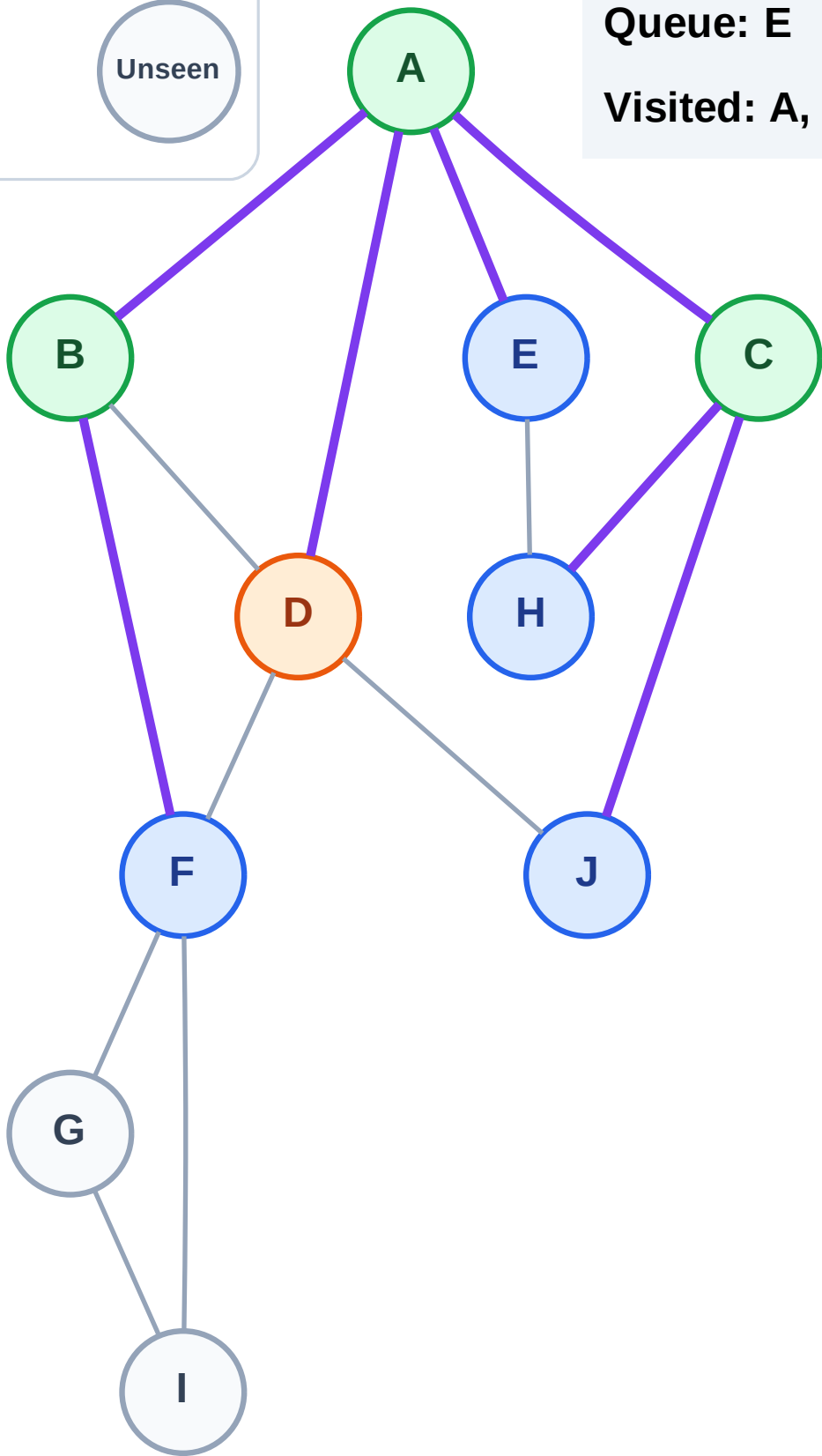
Processing

Discovered

Unseen

Queue: E → F → H → J

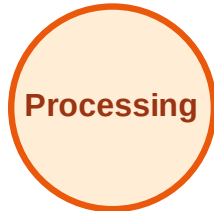
Visited: A, B, C



BFS Traversal

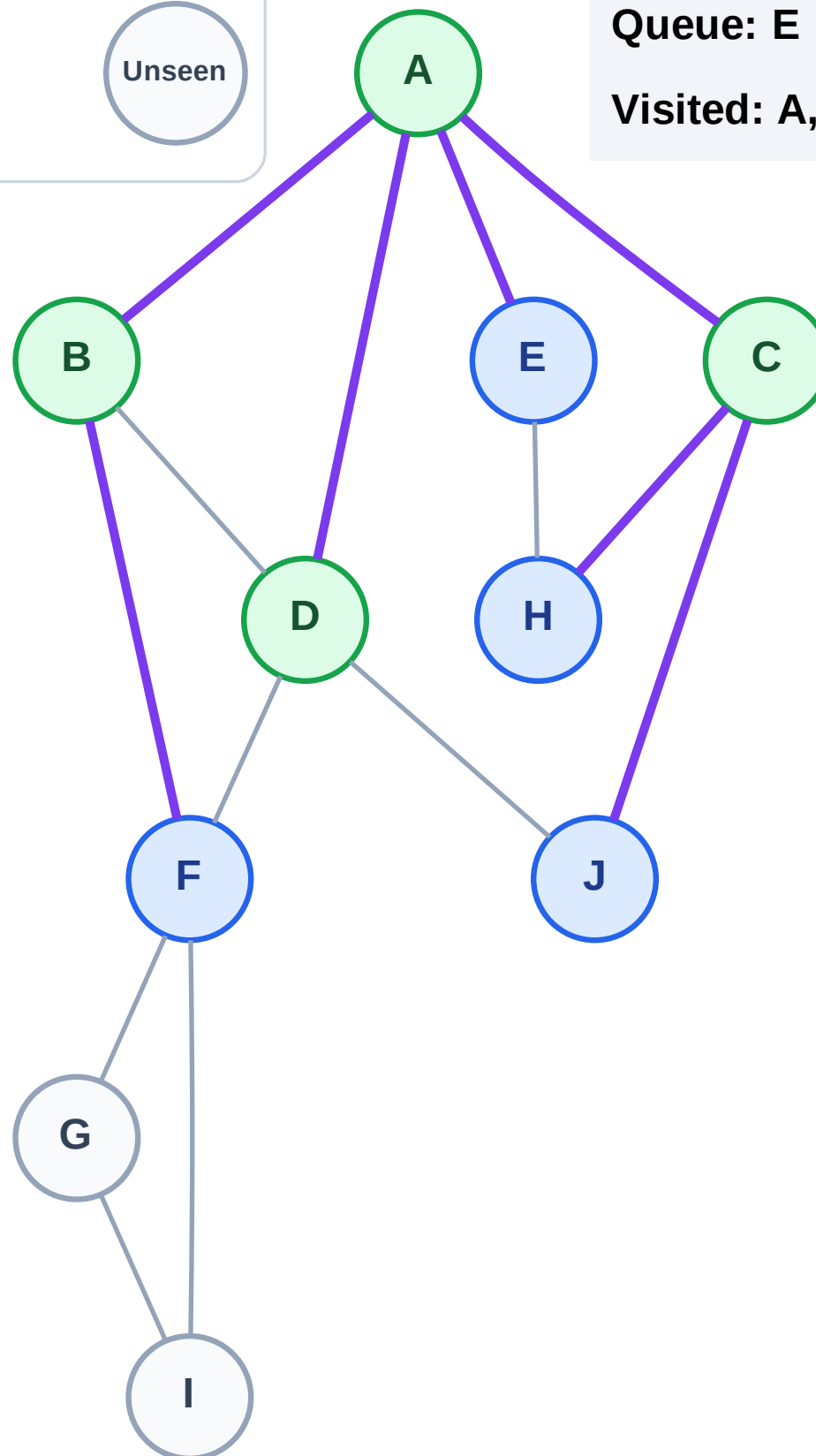
Step 9: No new neighbors from D

Legend



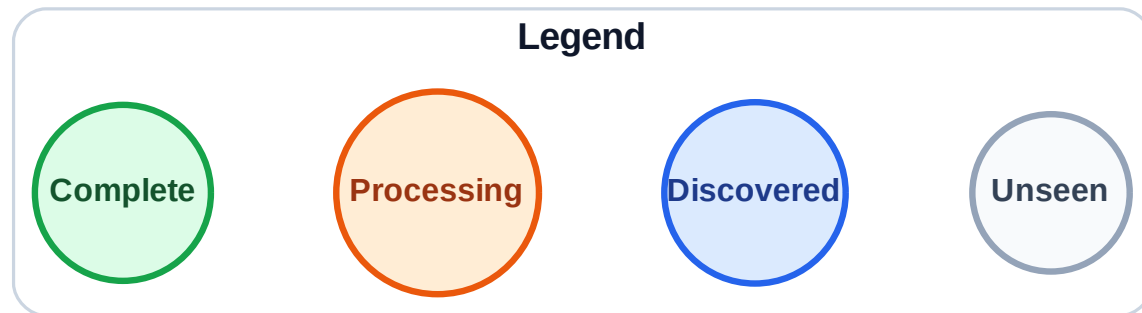
Queue: E → F → H → J

Visited: A, B, C, D

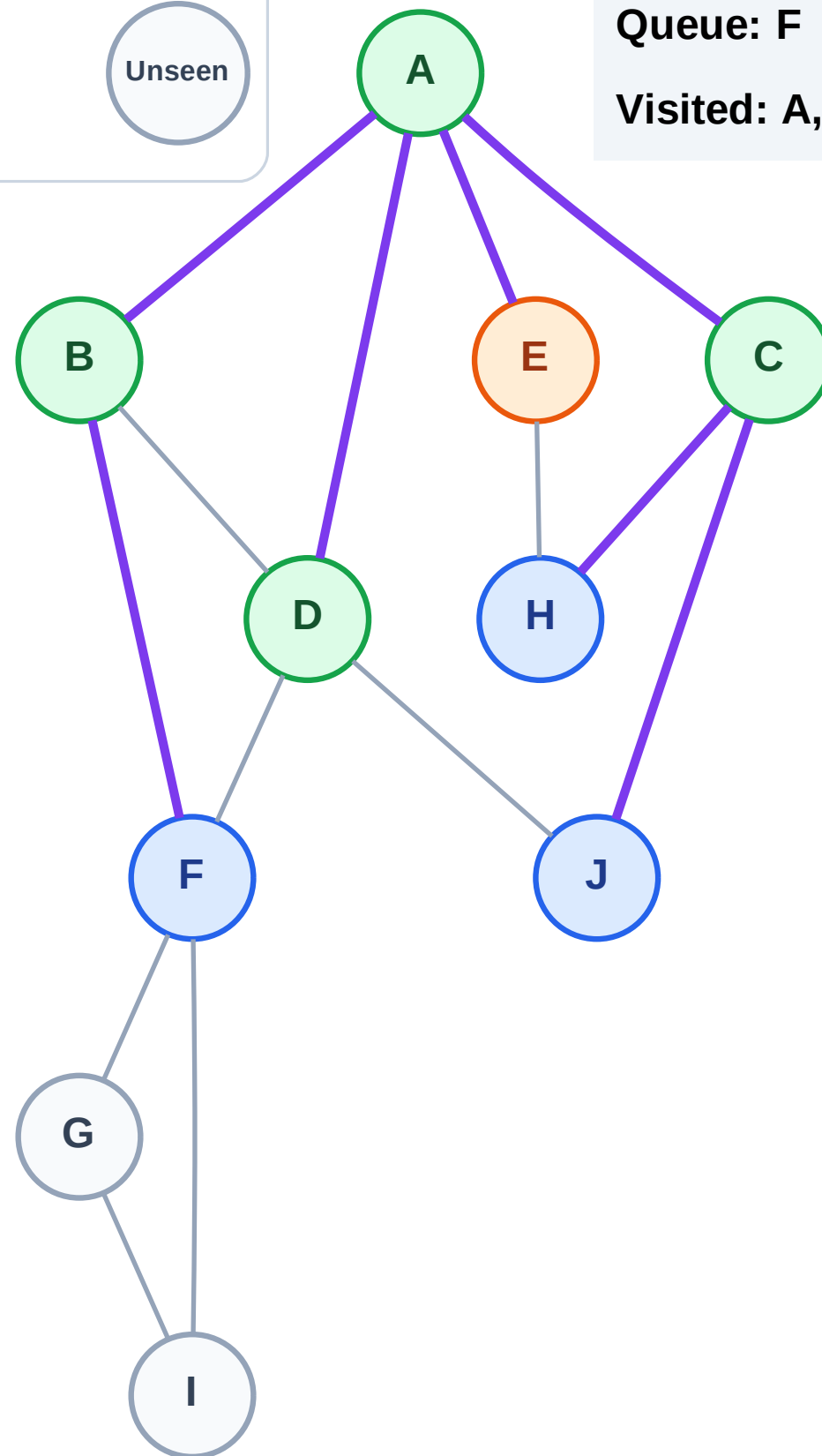


BFS Traversal

Step 10: Process node E



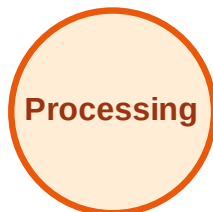
Queue: F → H → J
Visited: A, B, C, D



BFS Traversal

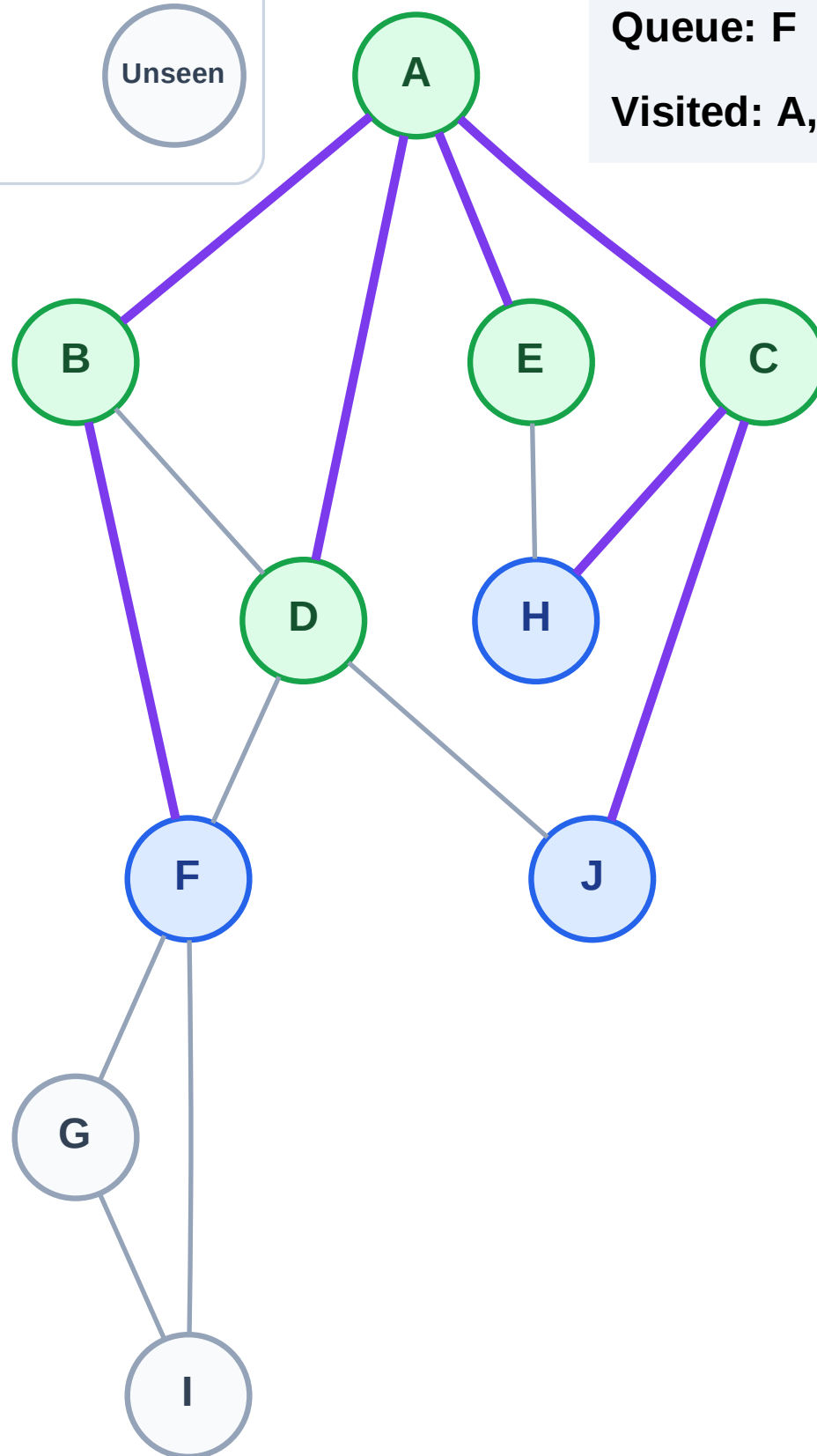
Step 11: No new neighbors from E

Legend



Queue: F → H → J

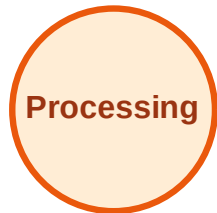
Visited: A, B, C, D, E



BFS Traversal

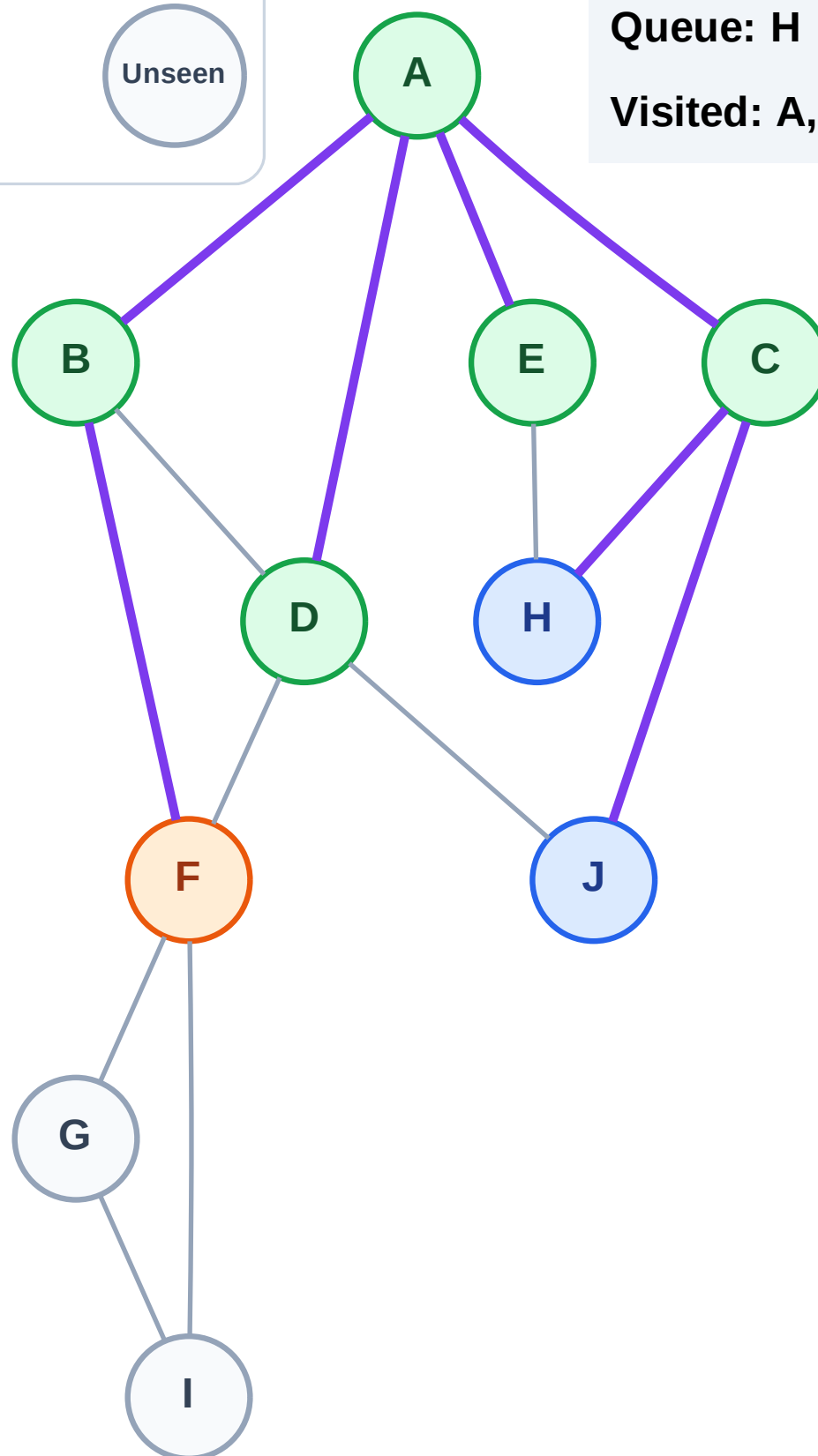
Step 12: Process node F

Legend



Queue: H → J

Visited: A, B, C, D, E



BFS Traversal

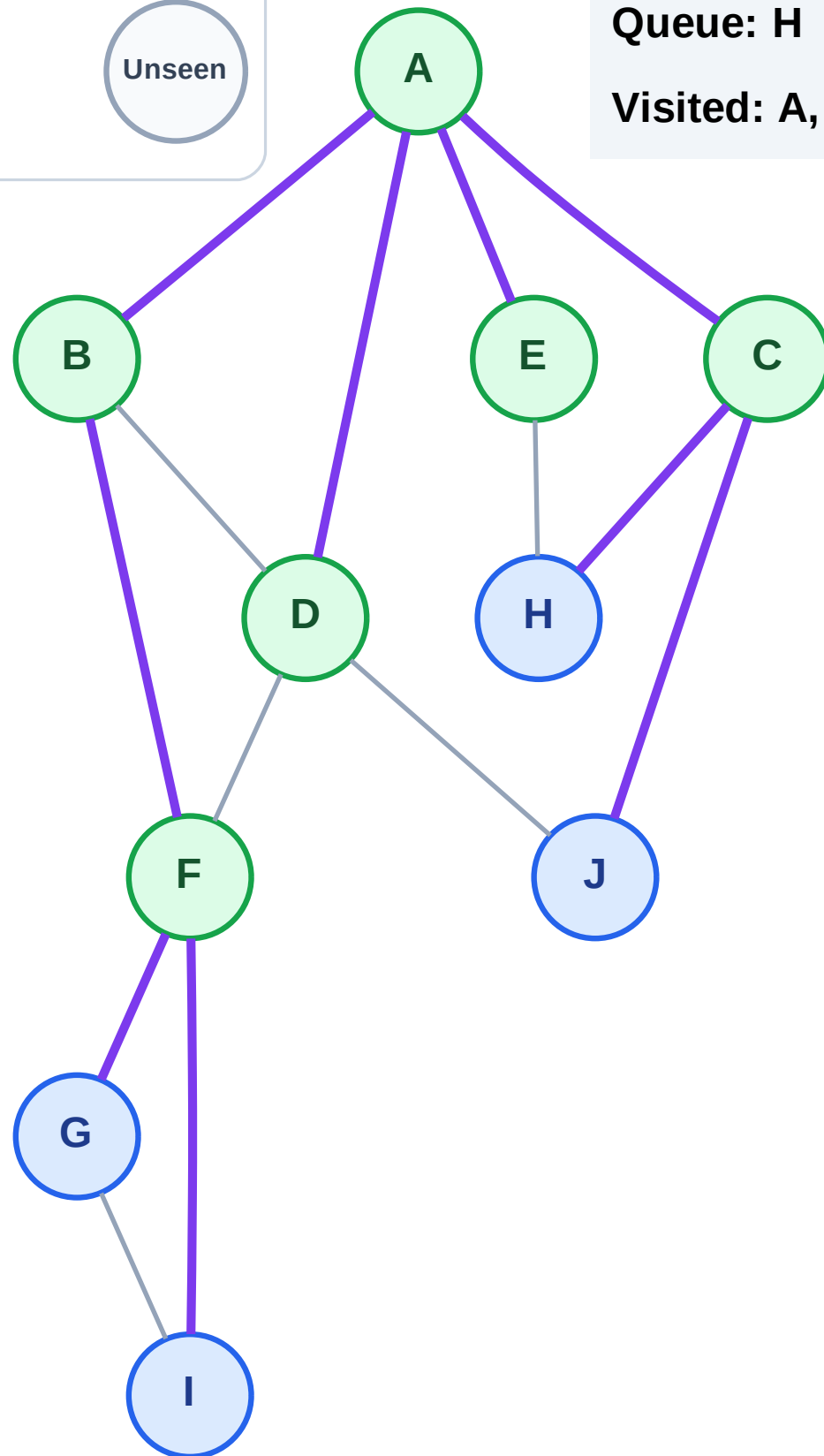
Step 13: Discover G, I from F

Legend



Queue: H → J → G → I

Visited: A, B, C, D, E, F



BFS Traversal

Step 14: Process node H

Legend

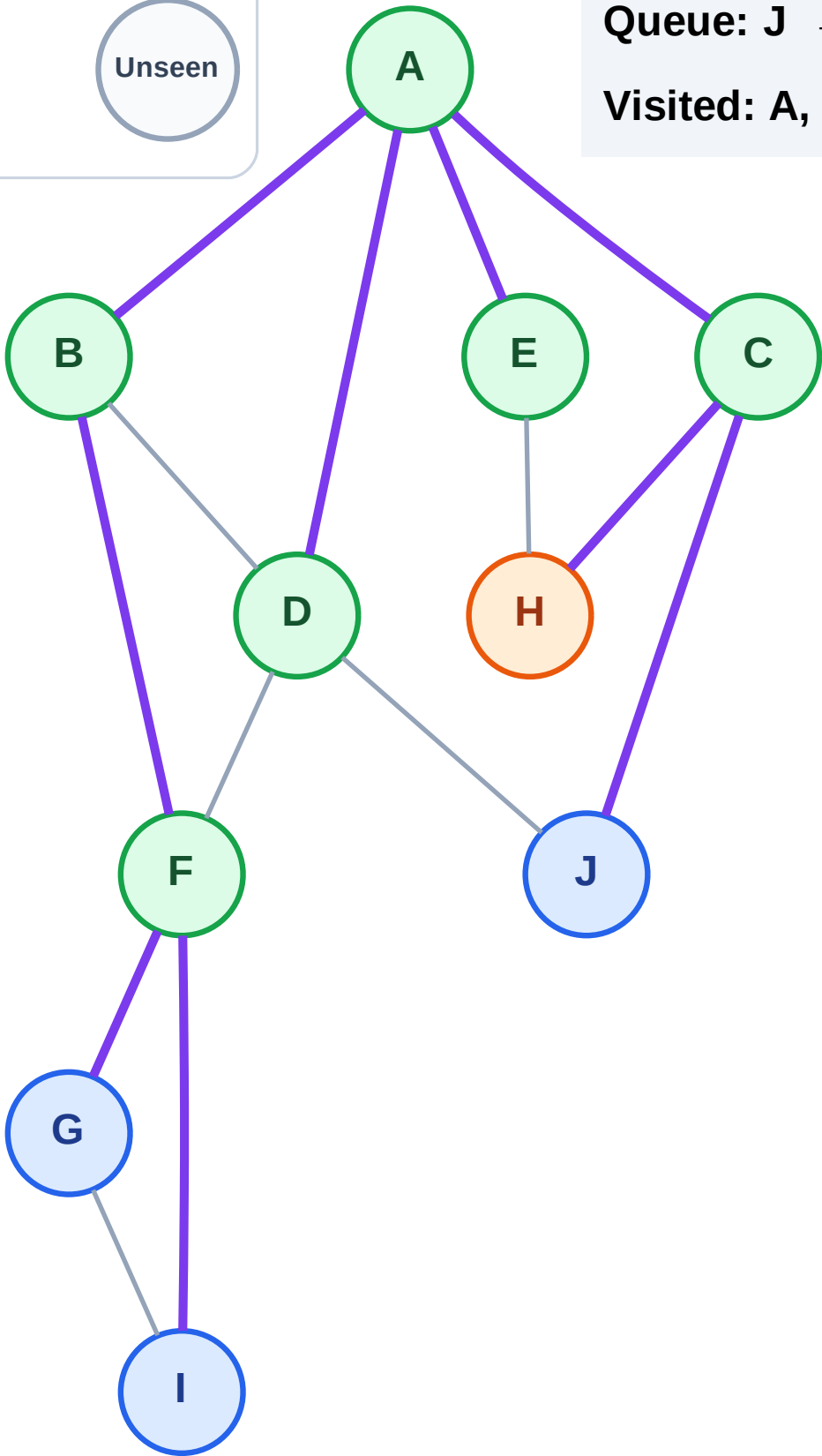
Complete

Processing

Discovered

Unseen

Queue: J → G → I
Visited: A, B, C, D, E, F



BFS Traversal

Step 15: No new neighbors from H

Legend

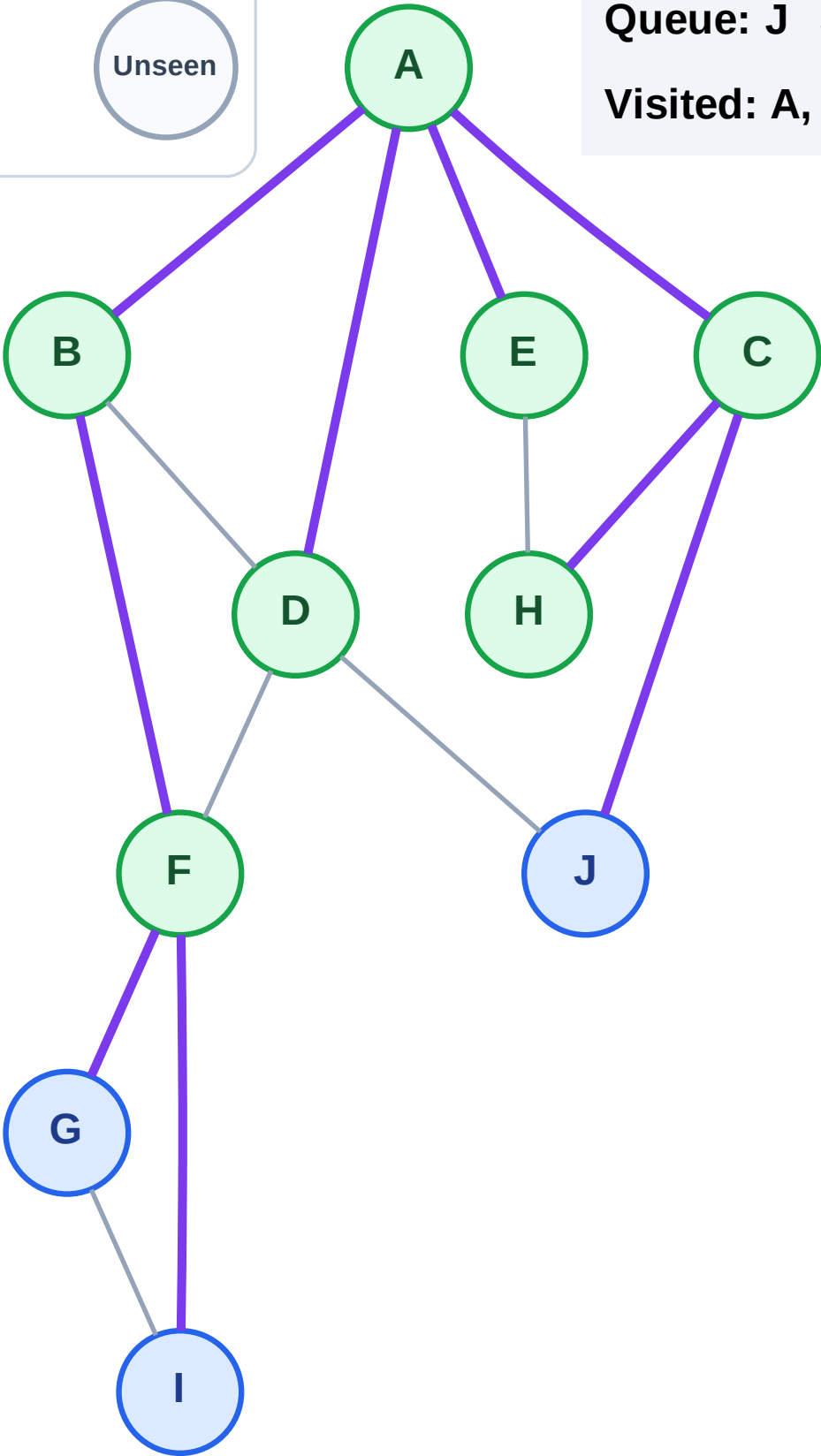
Complete

Processing

Discovered

Unseen

Queue: J → G → I
Visited: A, B, C, D, E, F, H



BFS Traversal

Step 16: Process node J

Legend

Complete

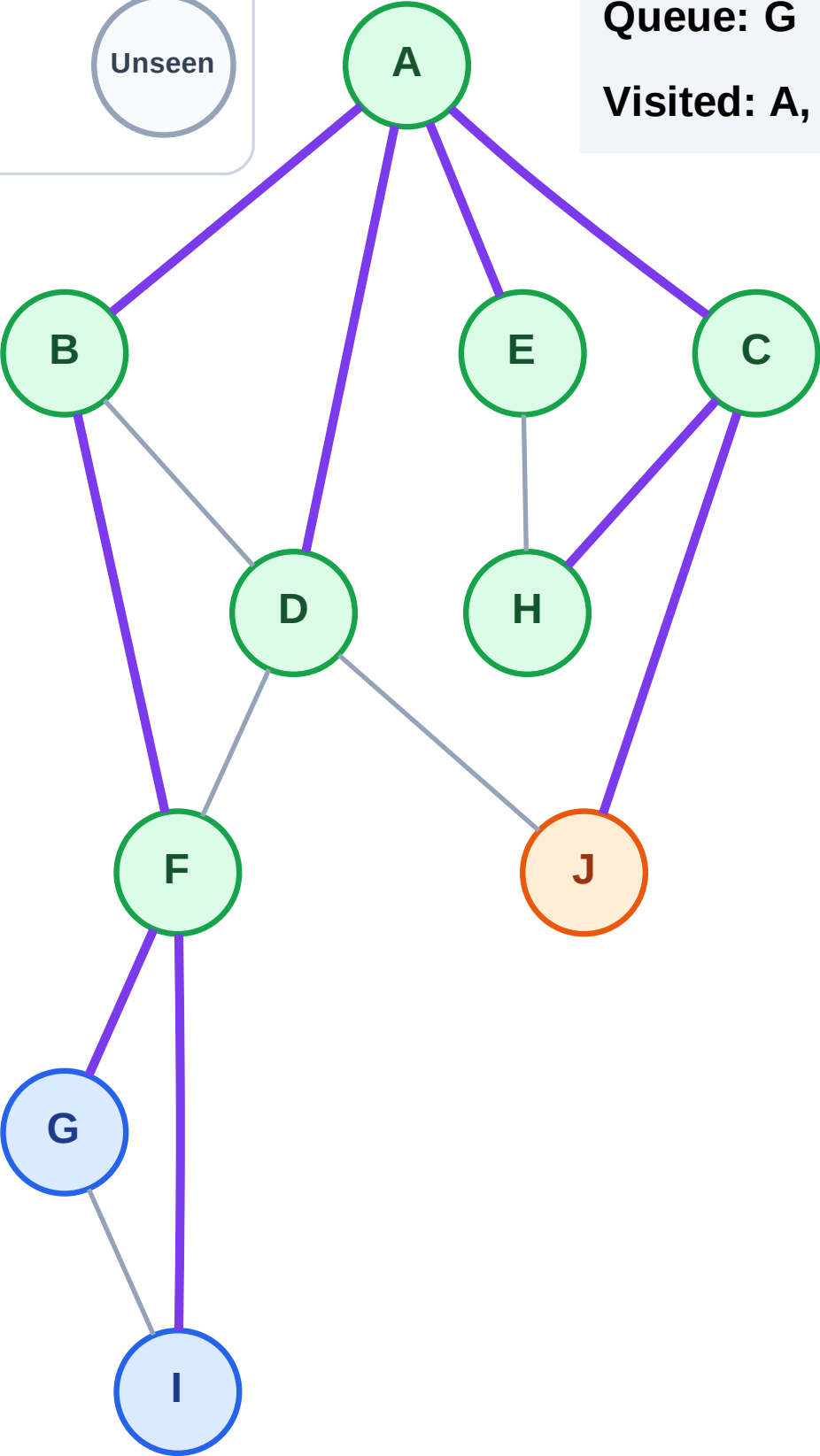
Processing

Discovered

Unseen

Queue: G → I

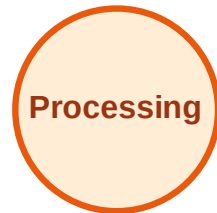
Visited: A, B, C, D, E, F, H



BFS Traversal

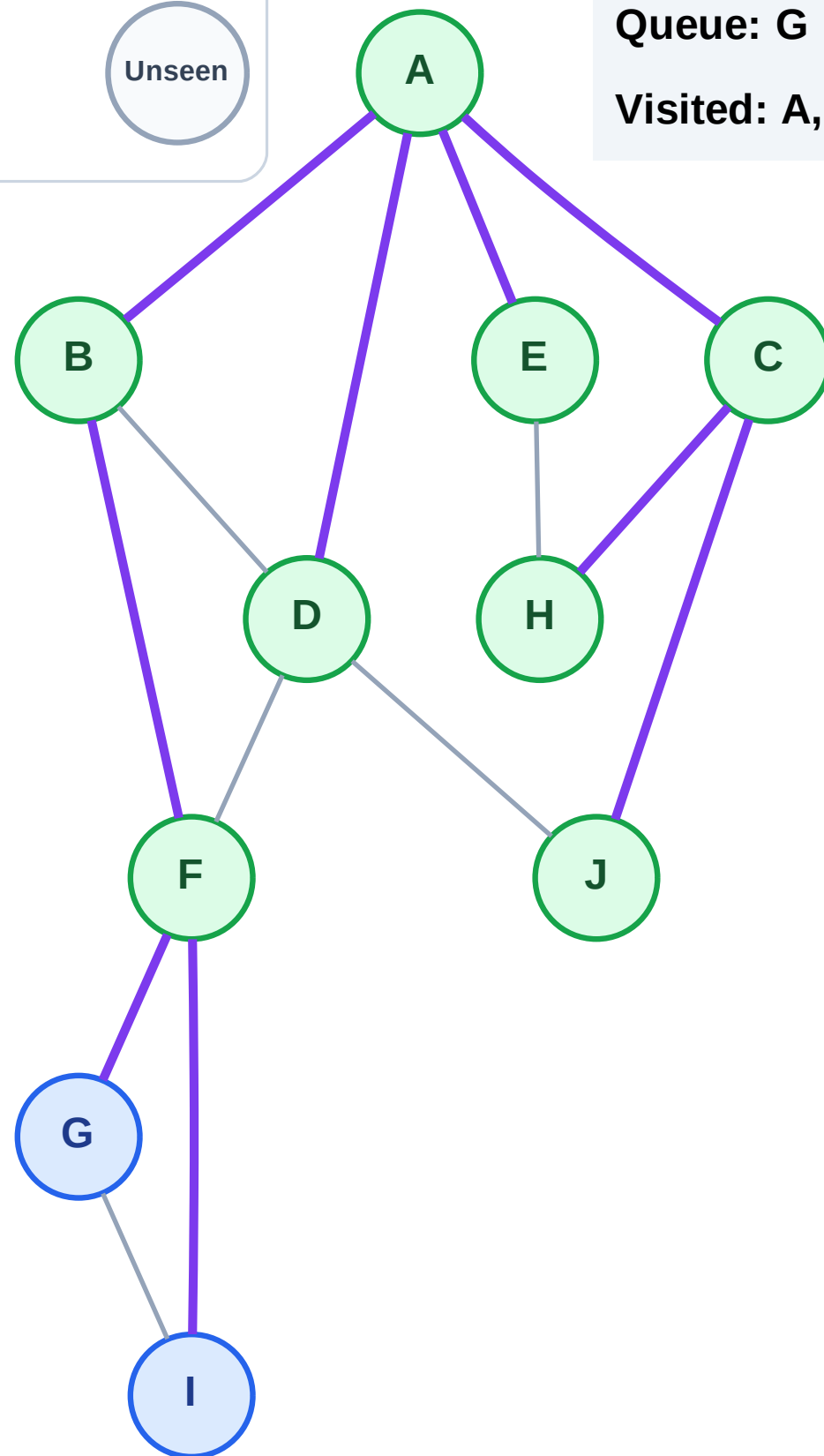
Step 17: No new neighbors from J

Legend



Queue: G → I

Visited: A, B, C, D, E, F, H, J



BFS Traversal

Step 18: Process node G

Legend

Complete

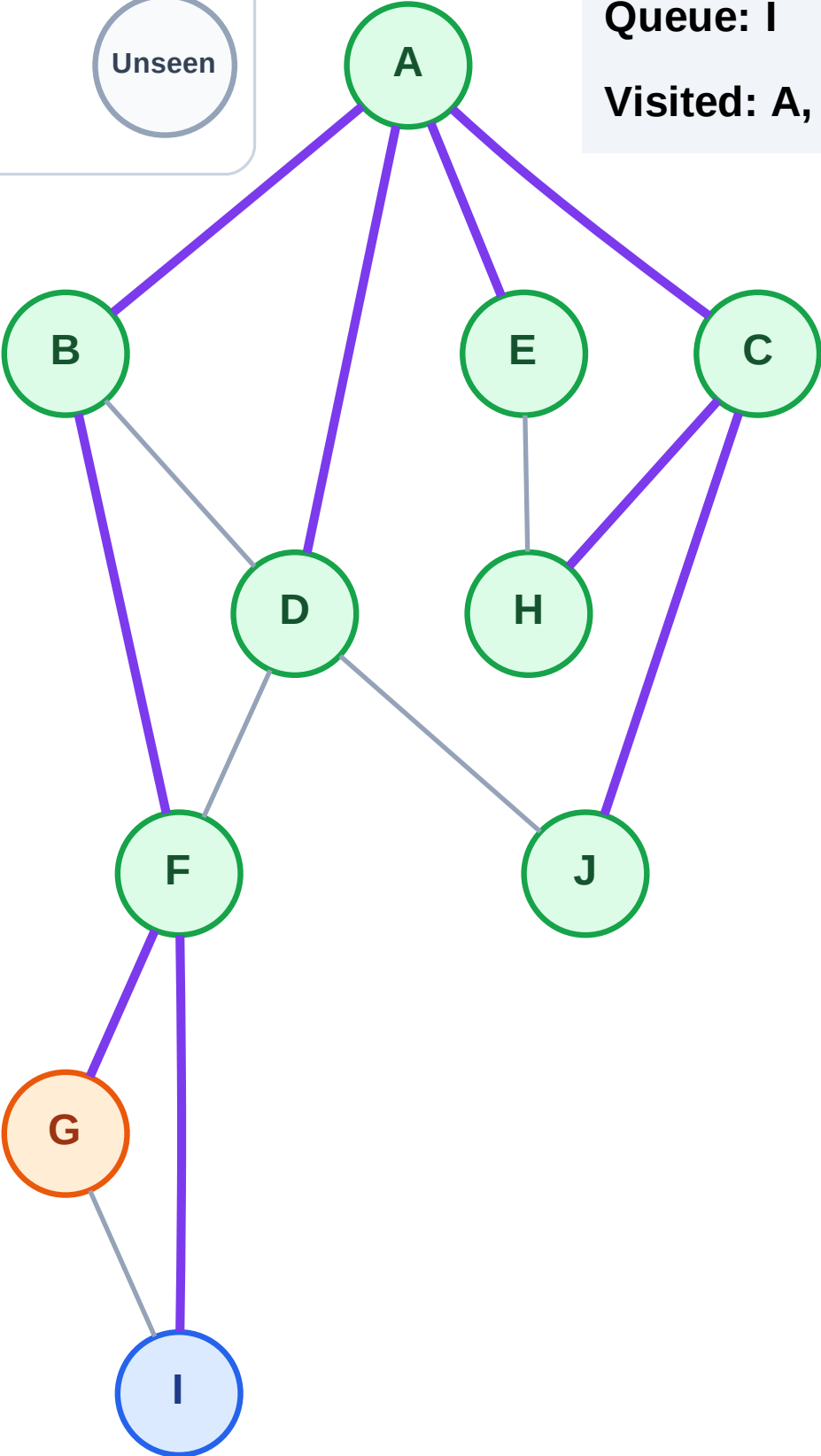
Processing

Discovered

Unseen

Queue: I

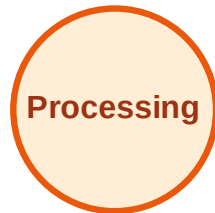
Visited: A, B, C, D, E, F, H, J



BFS Traversal

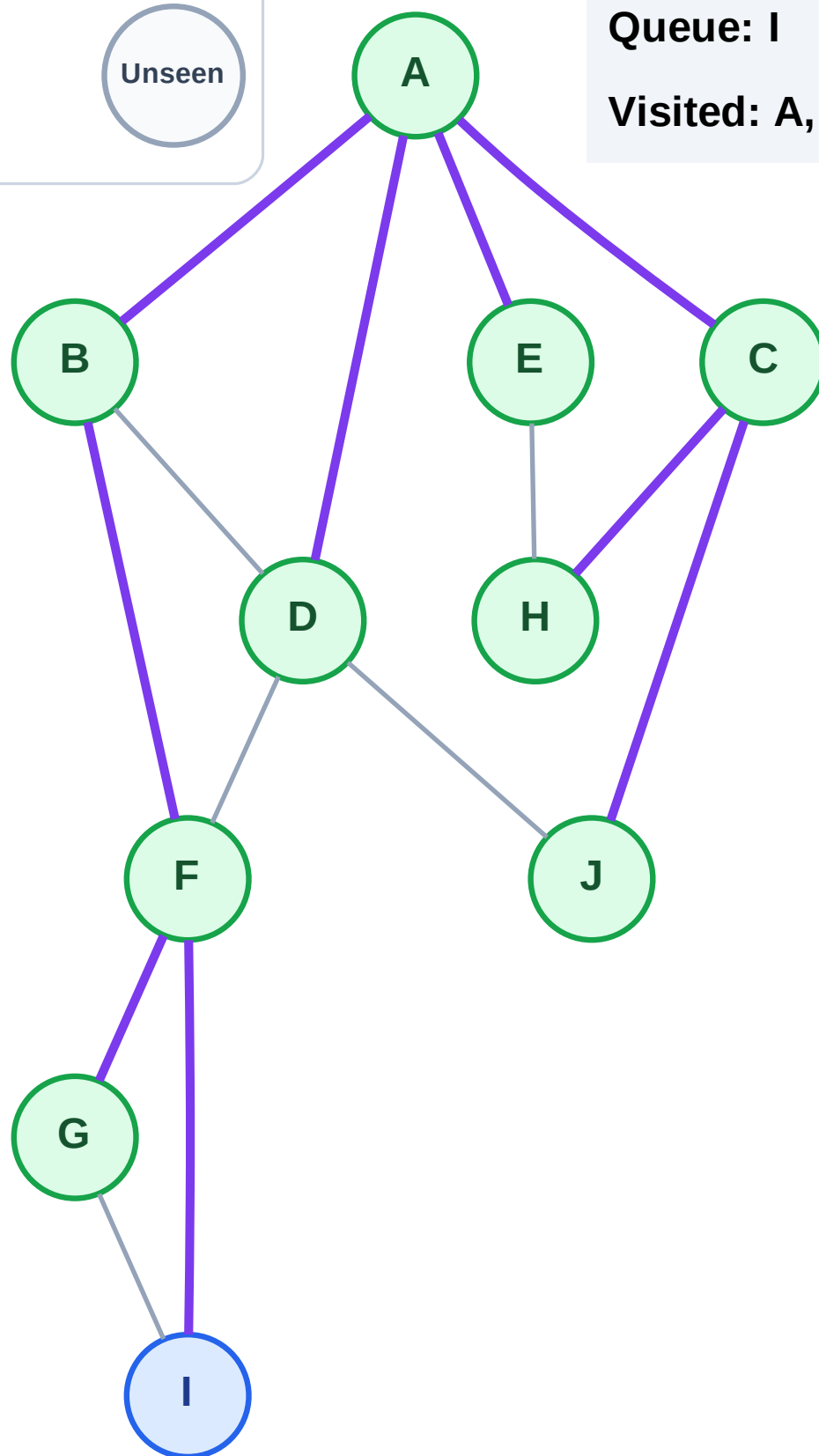
Step 19: No new neighbors from G

Legend



Queue: I

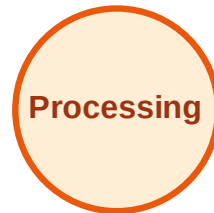
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

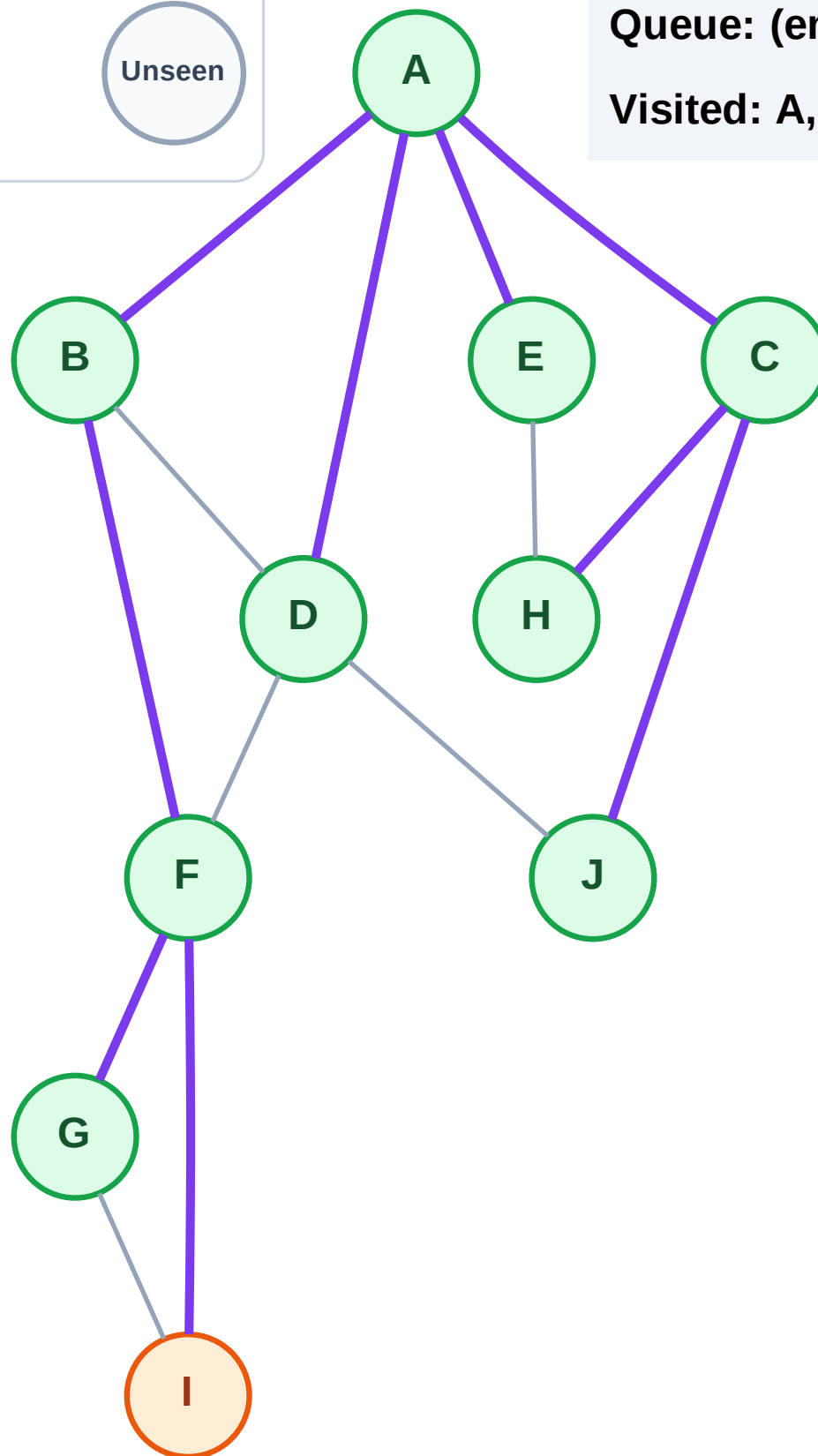
Step 20: Process node I

Legend



Queue: (empty)

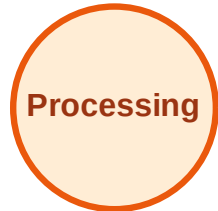
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

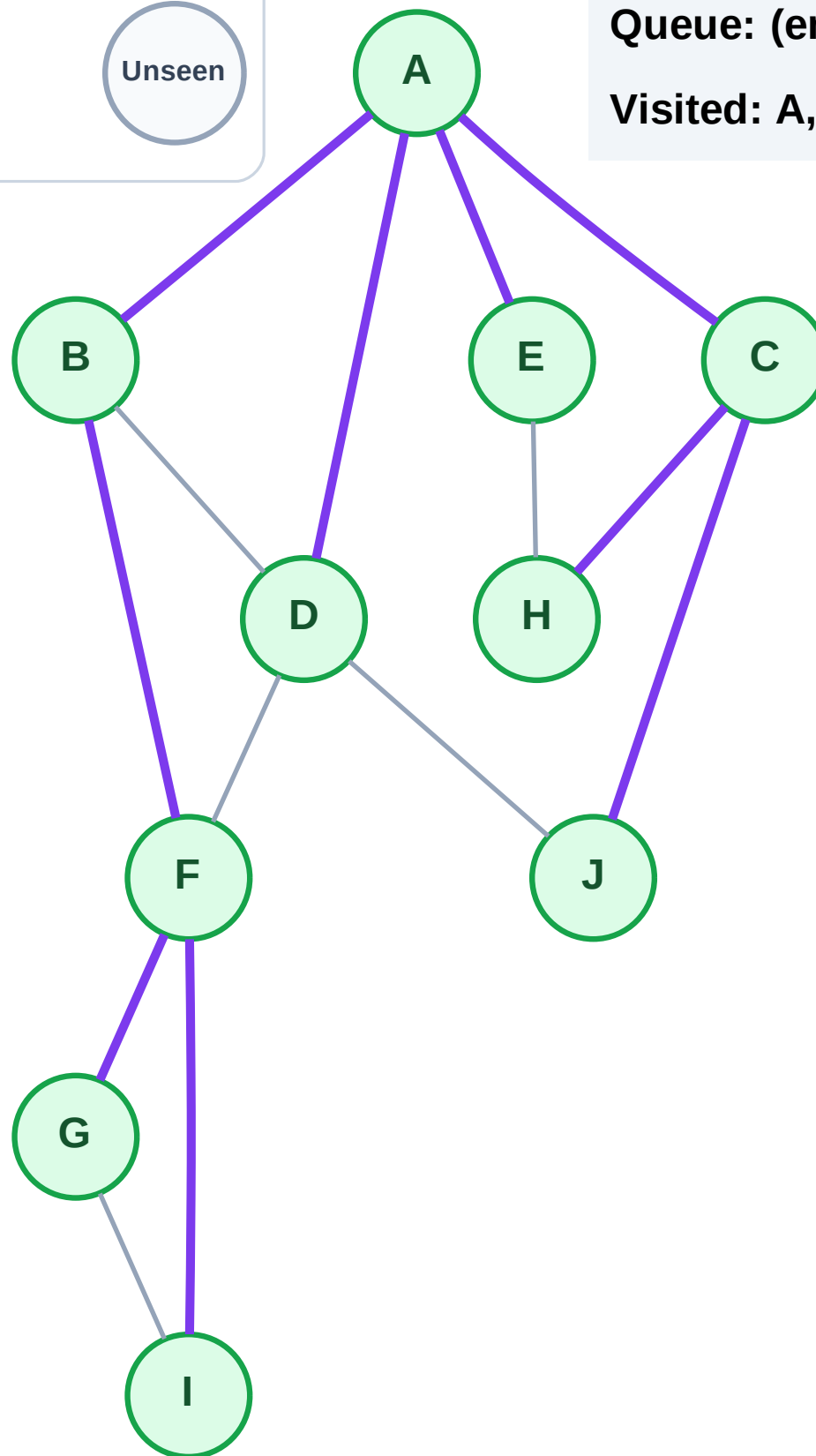
Step 21: No new neighbors from I

Legend



Queue: (empty)

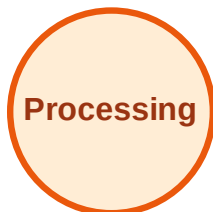
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

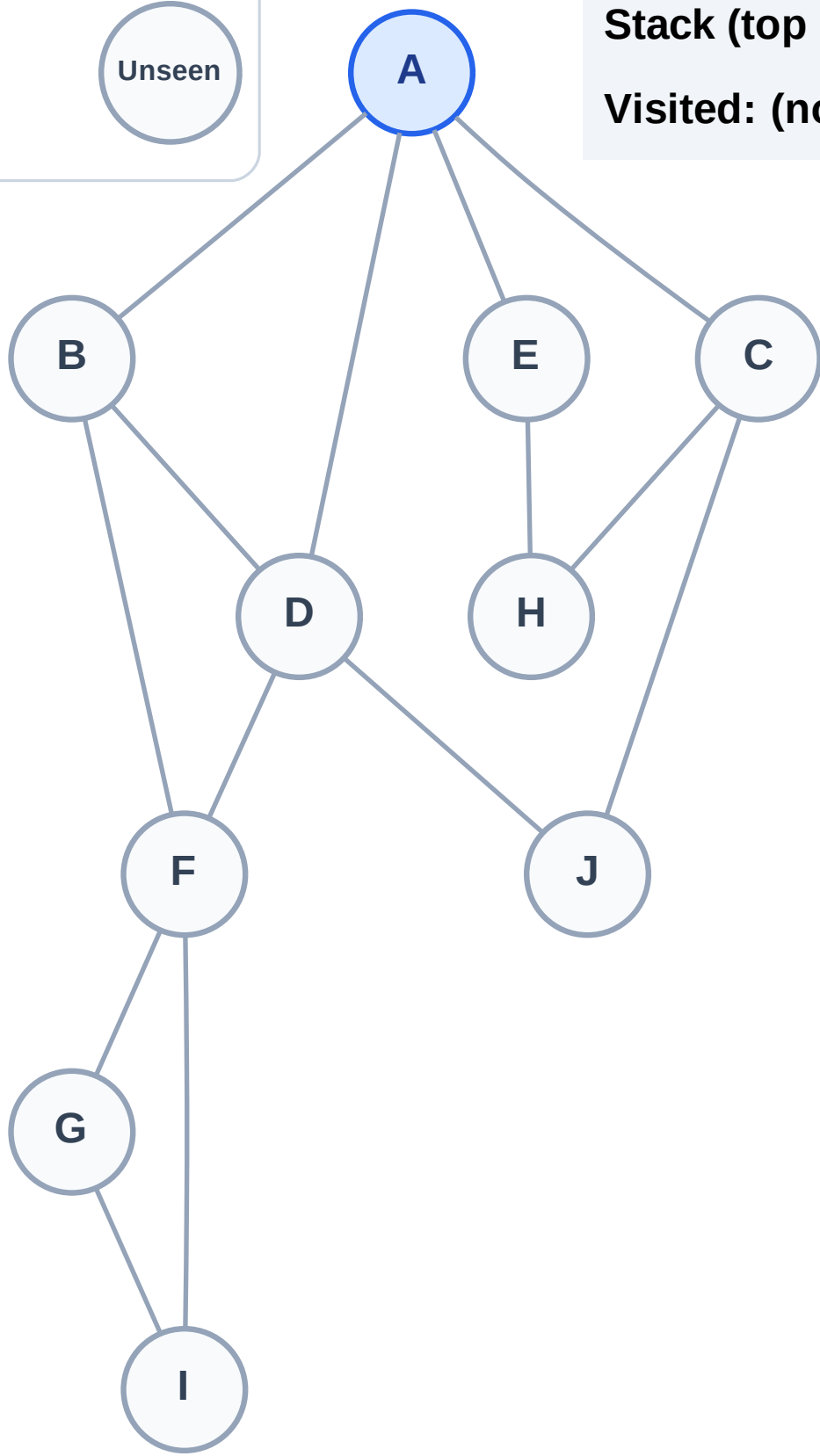
Step 1: Start at A

Legend



Stack (top at right): A

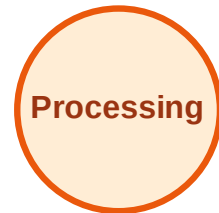
Visited: (none)



DFS Traversal

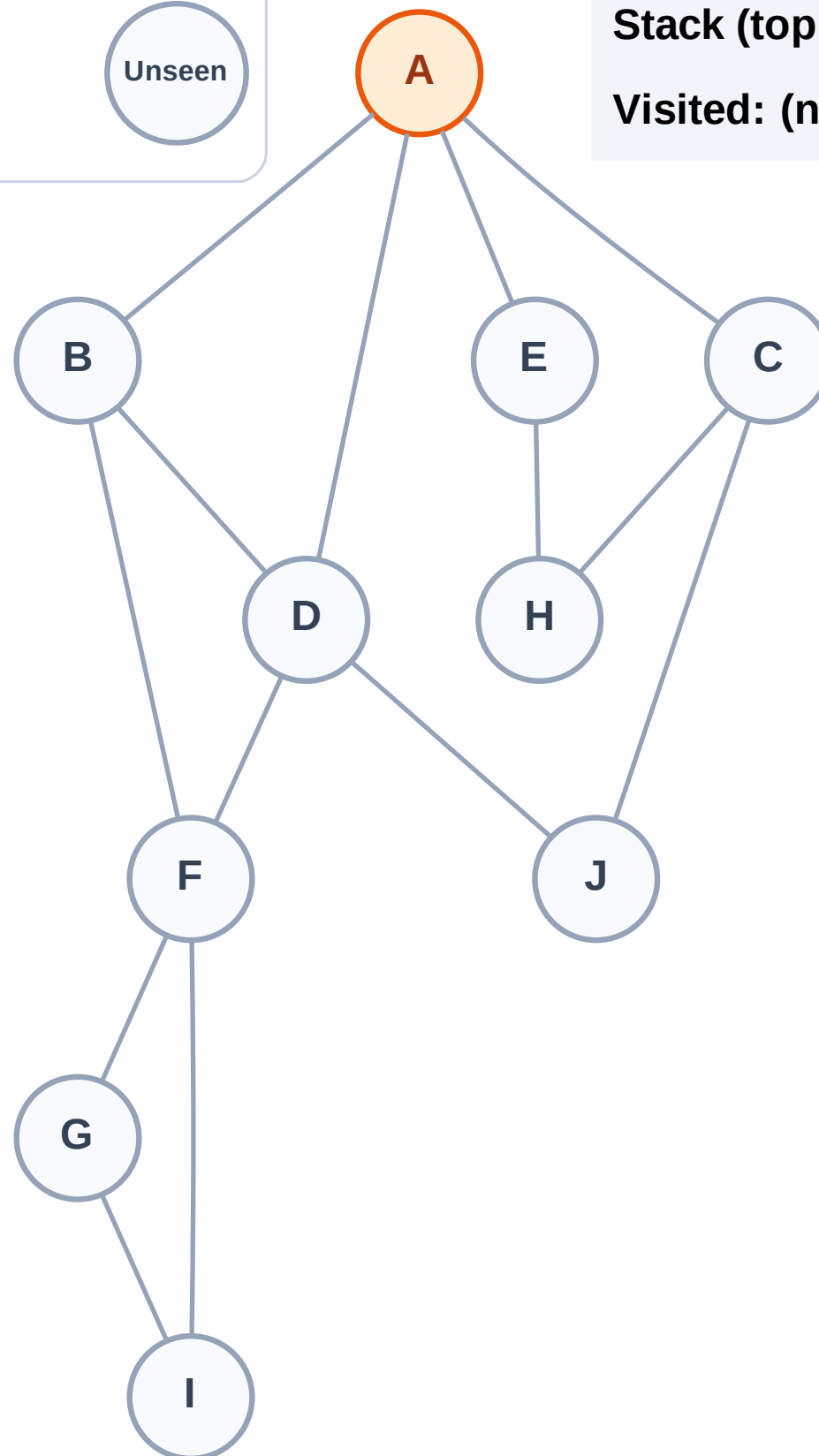
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover E, D, C, B from A

Legend

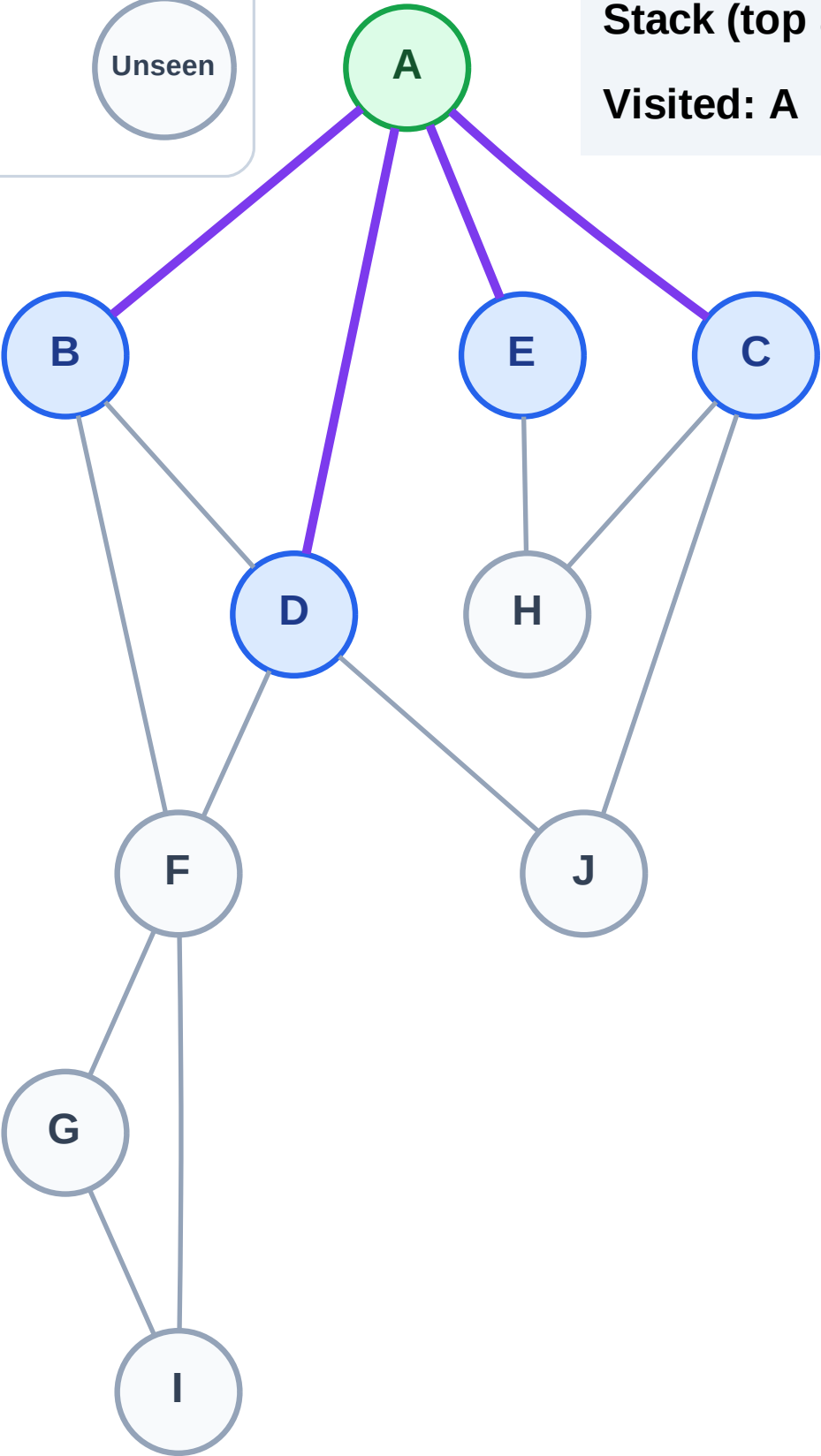
Complete

Processing

Discovered

Unseen

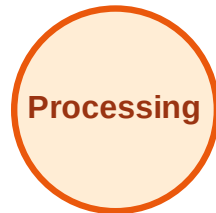
Stack (top at right): E | D | C | B
Visited: A



DFS Traversal

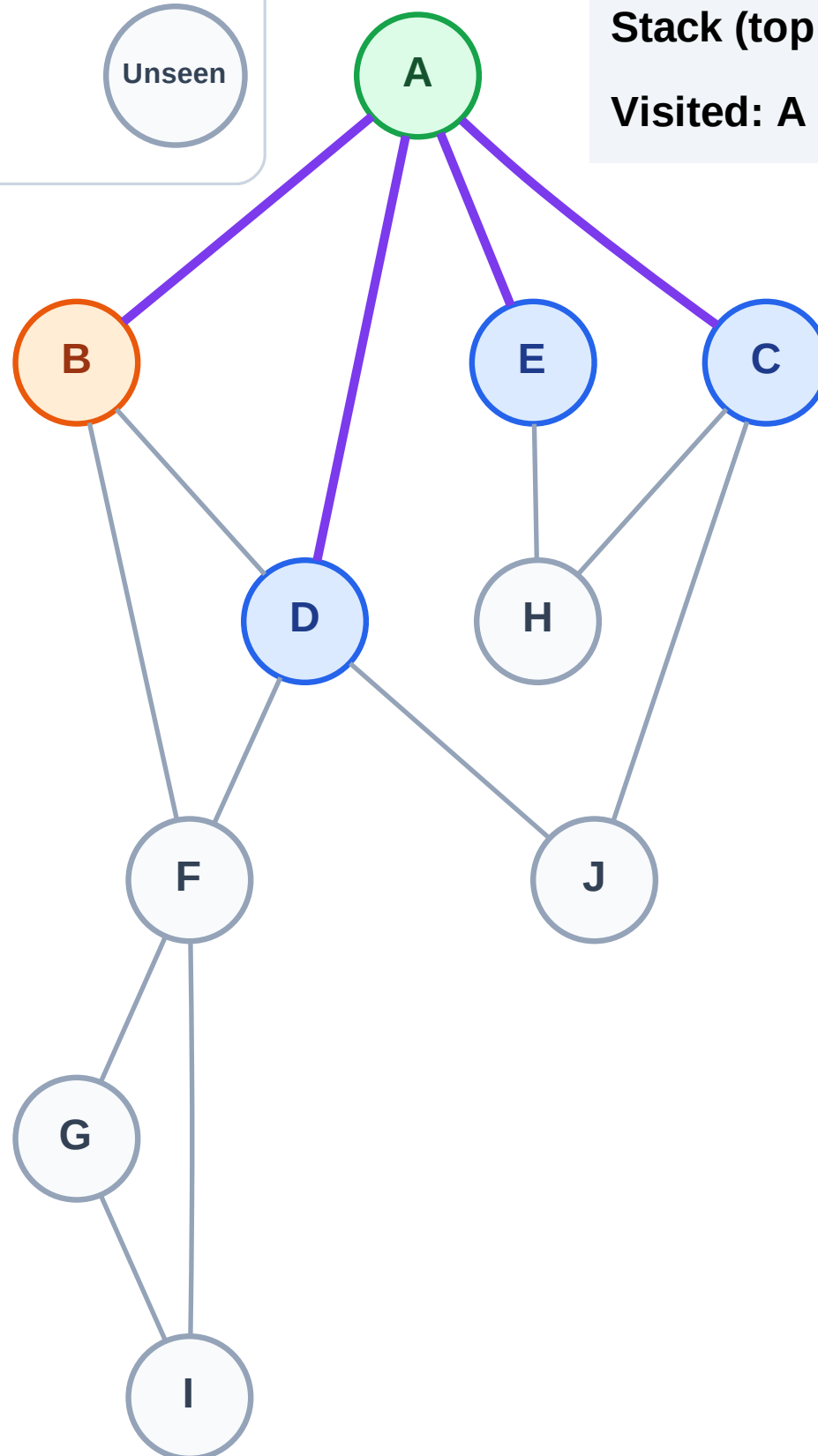
Step 4: Process node B

Legend



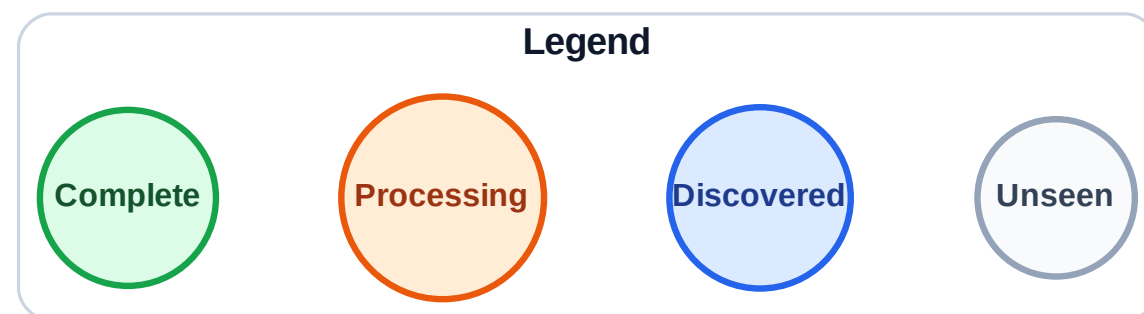
Stack (top at right): E | D | C

Visited: A

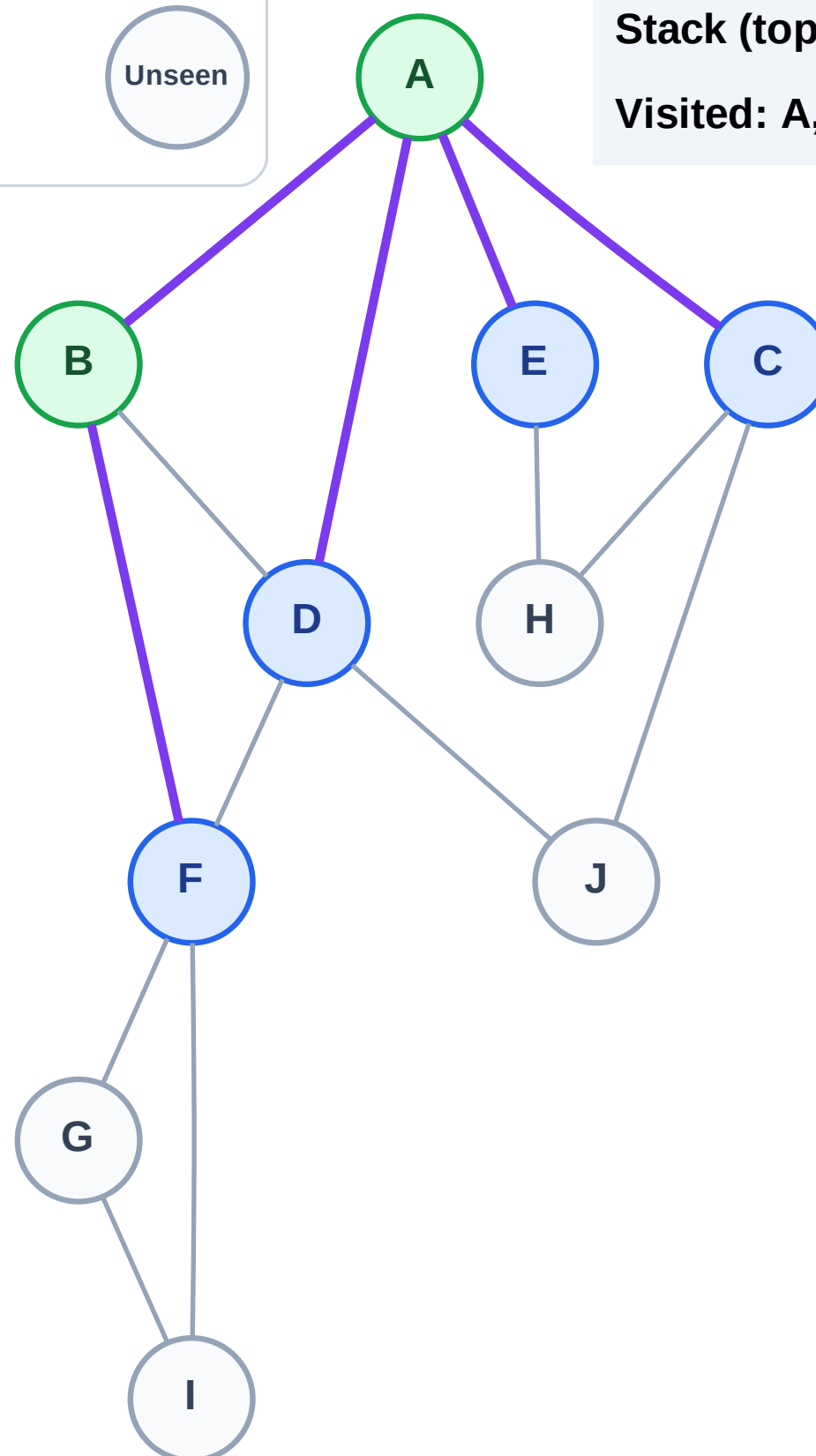


DFS Traversal

Step 5: Discover F from B



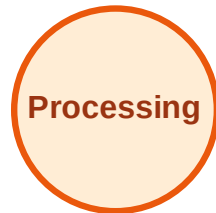
Stack (top at right): E | D | C | F
Visited: A, B



DFS Traversal

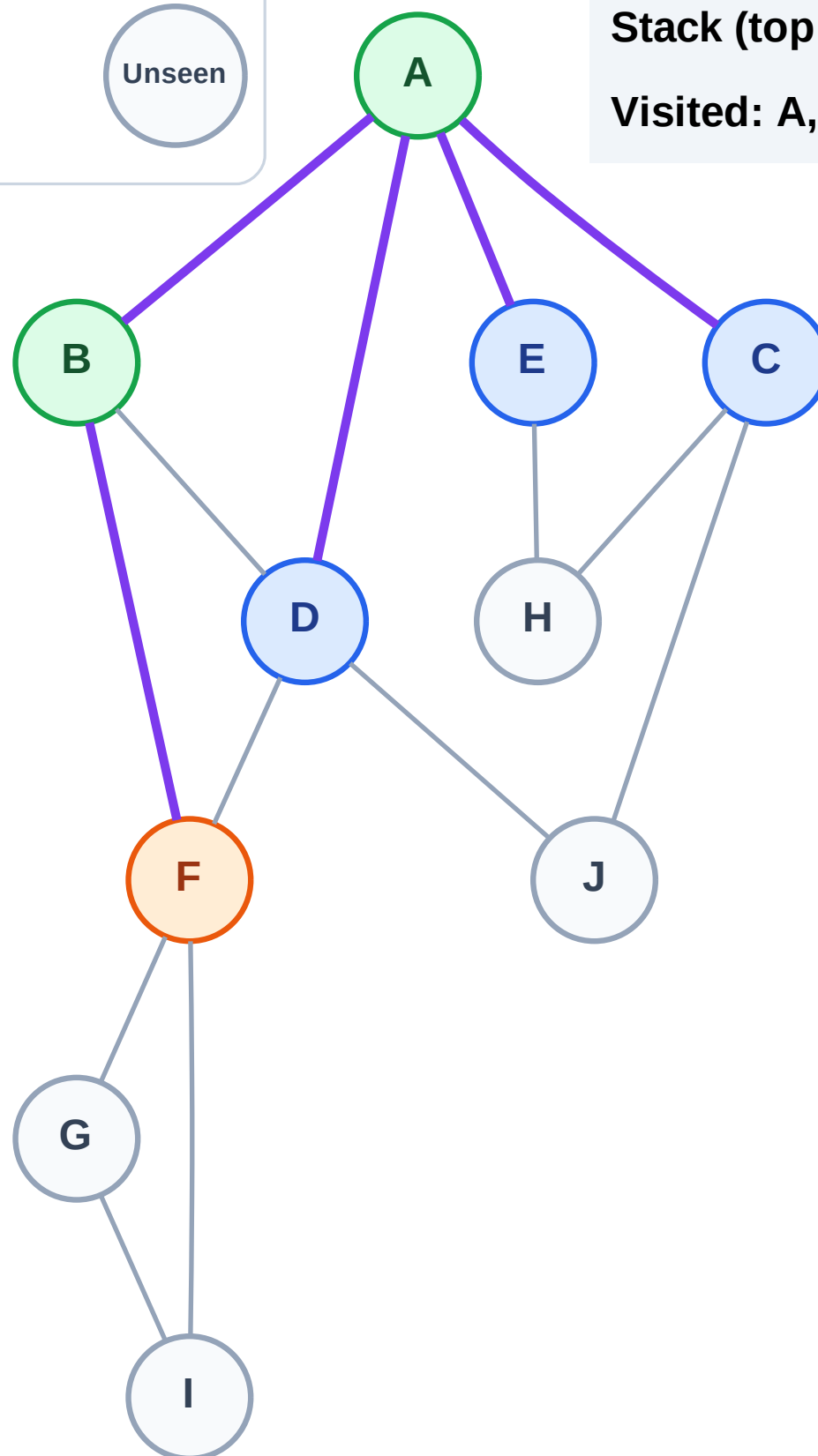
Step 6: Process node F

Legend



Stack (top at right): E | D | C

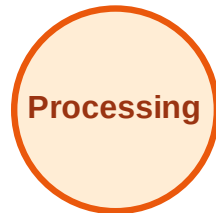
Visited: A, B



DFS Traversal

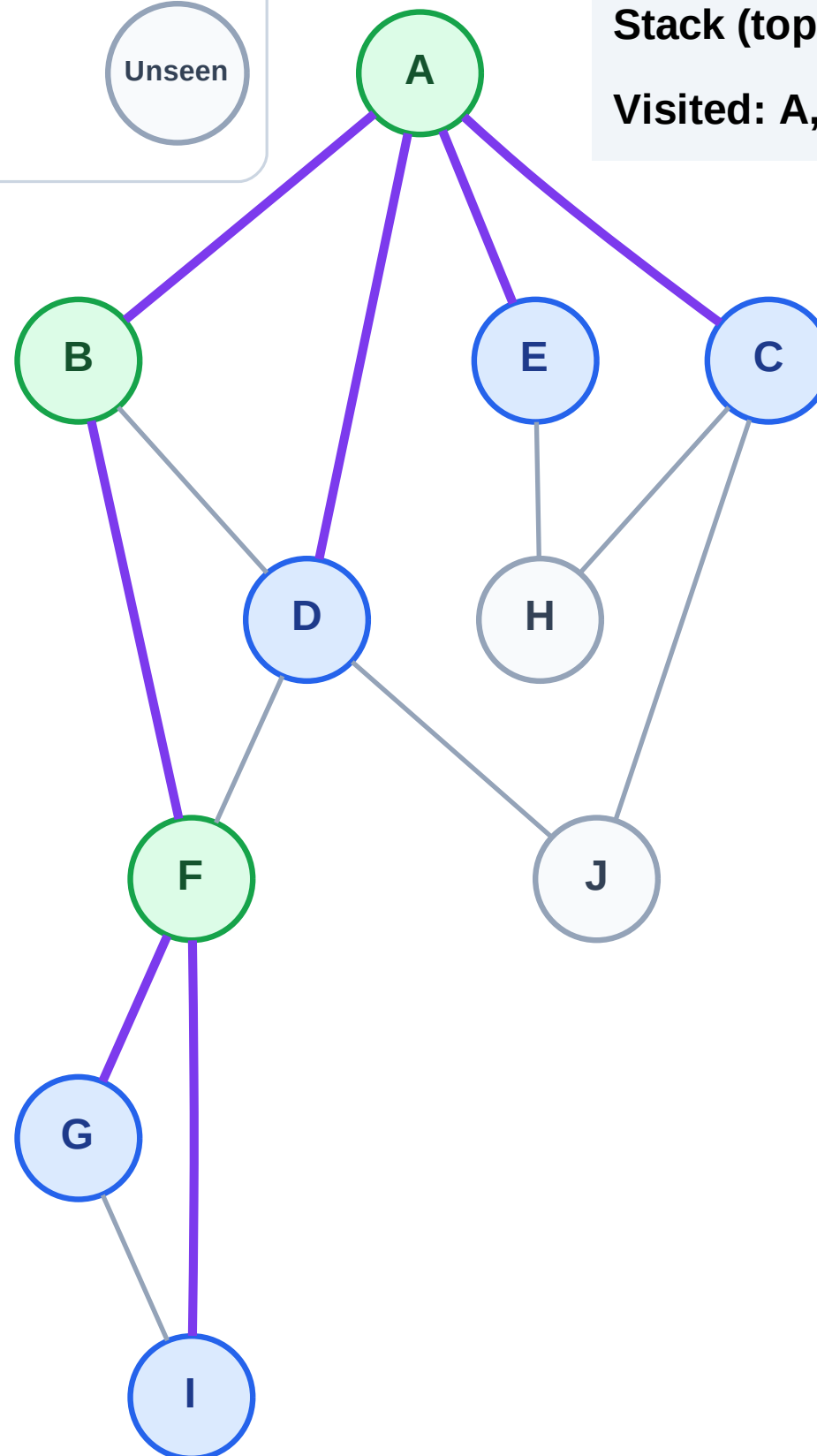
Step 7: Discover I, G from F

Legend



Stack (top at right): E | D | C | I | G

Visited: A, B, F



DFS Traversal

Step 8: Process node G

Legend

Complete

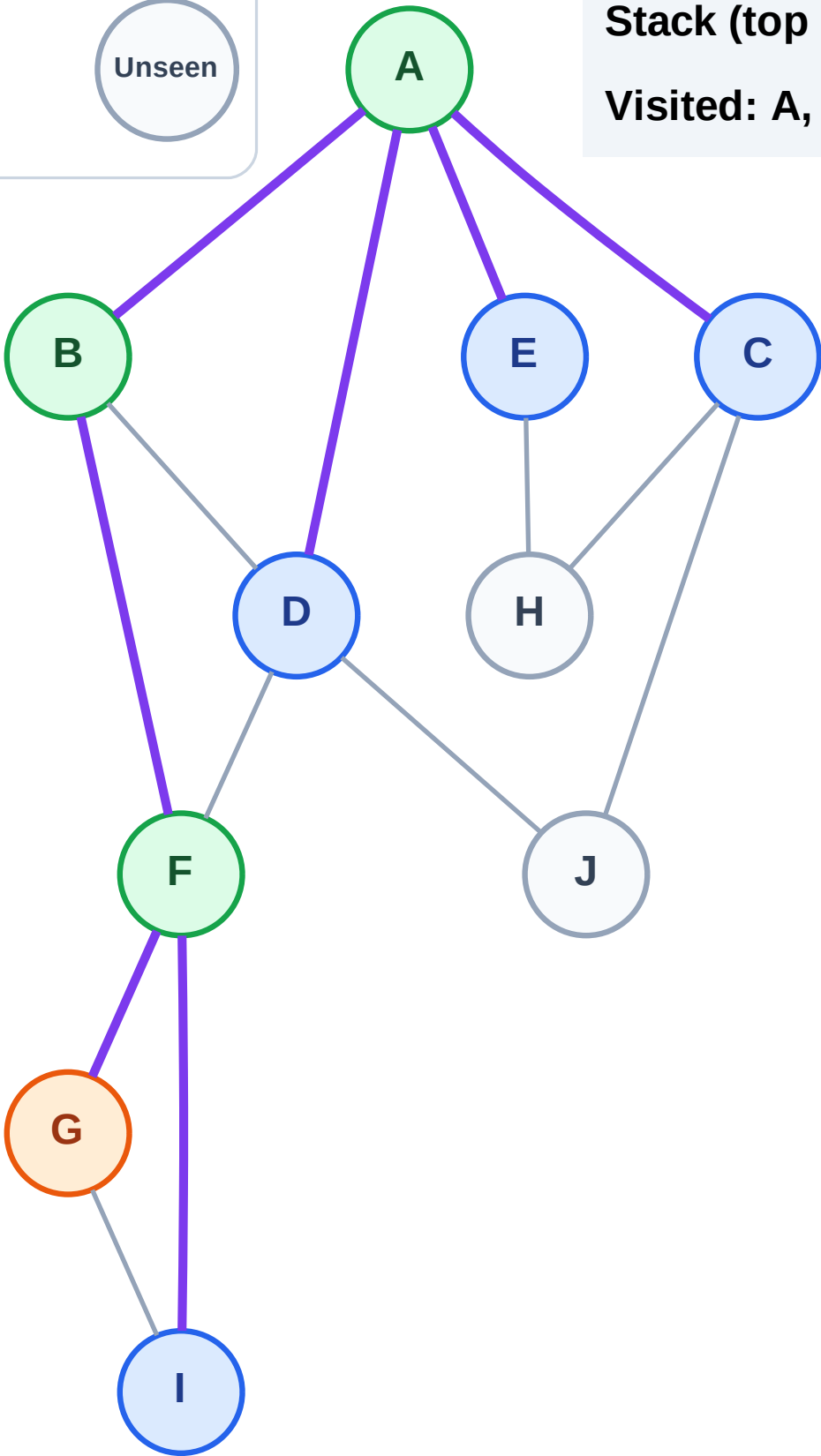
Processing

Discovered

Unseen

Stack (top at right): E | D | C | I

Visited: A, B, F



DFS Traversal

Step 9: No new neighbors from G

Legend

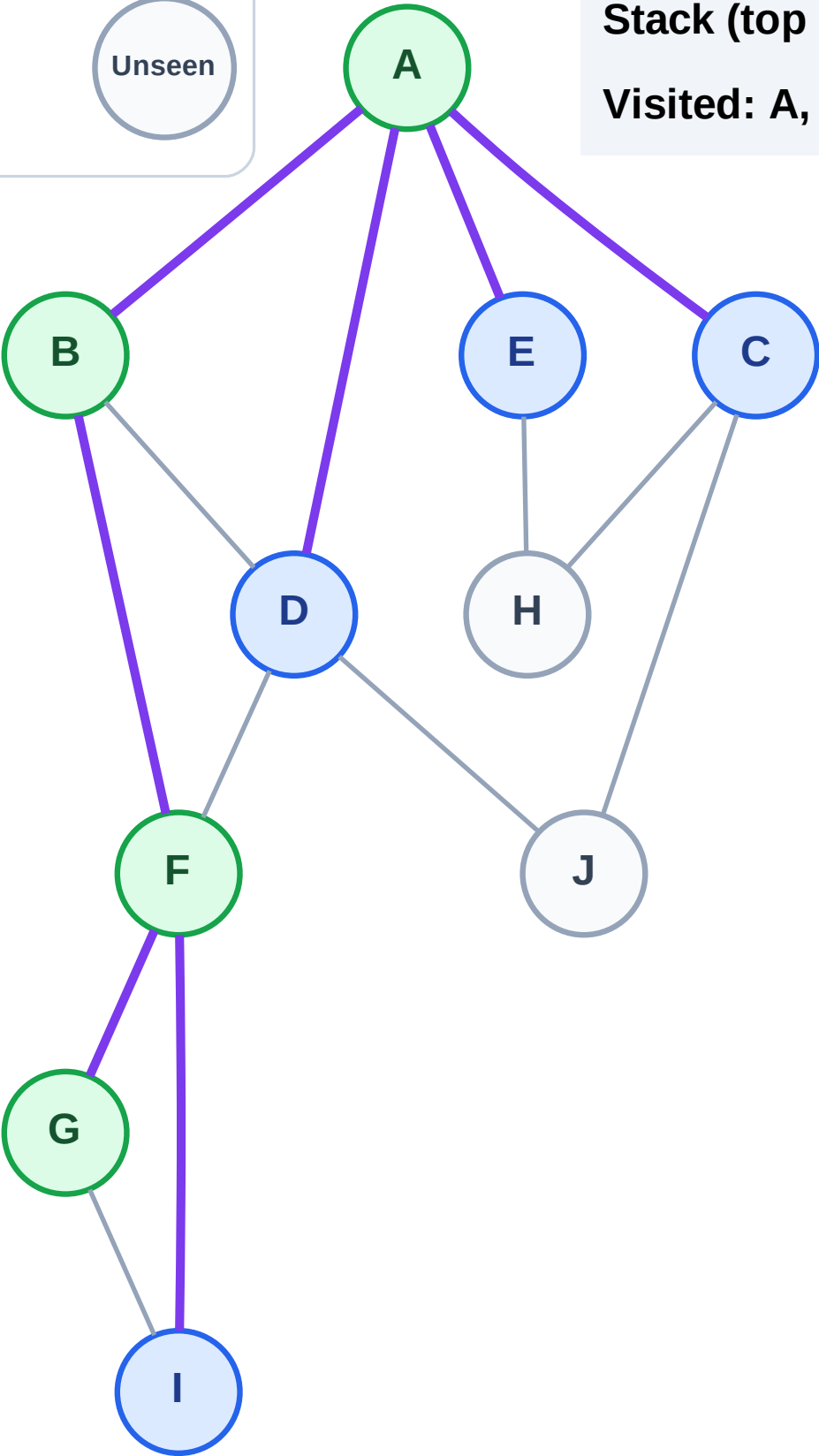
Complete

Processing

Discovered

Unseen

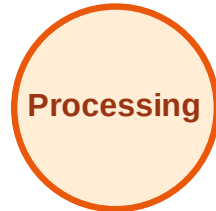
Stack (top at right): E | D | C | I
Visited: A, B, F, G



DFS Traversal

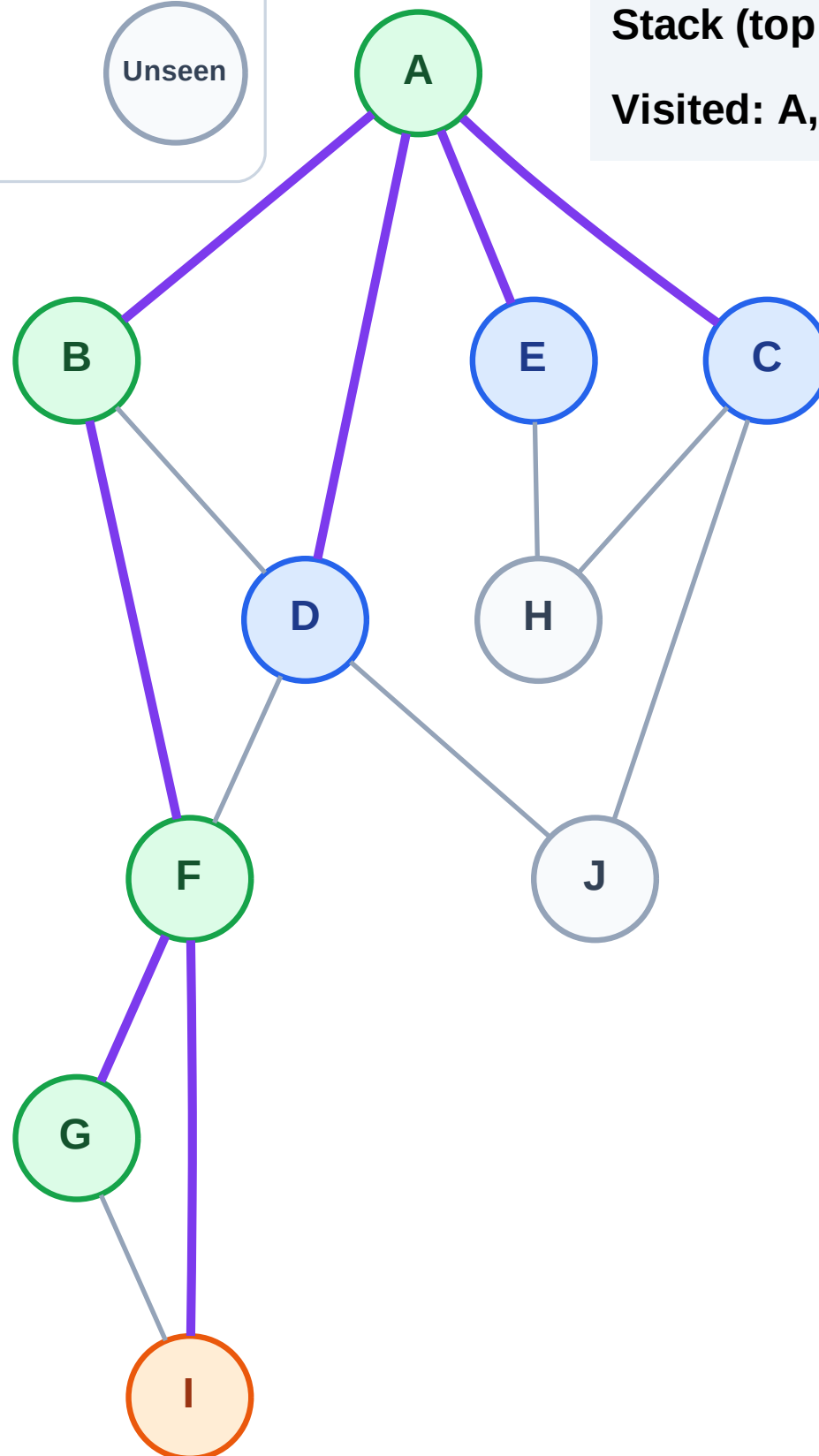
Step 10: Process node I

Legend



Stack (top at right): E | D | C

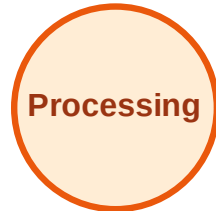
Visited: A, B, F, G



DFS Traversal

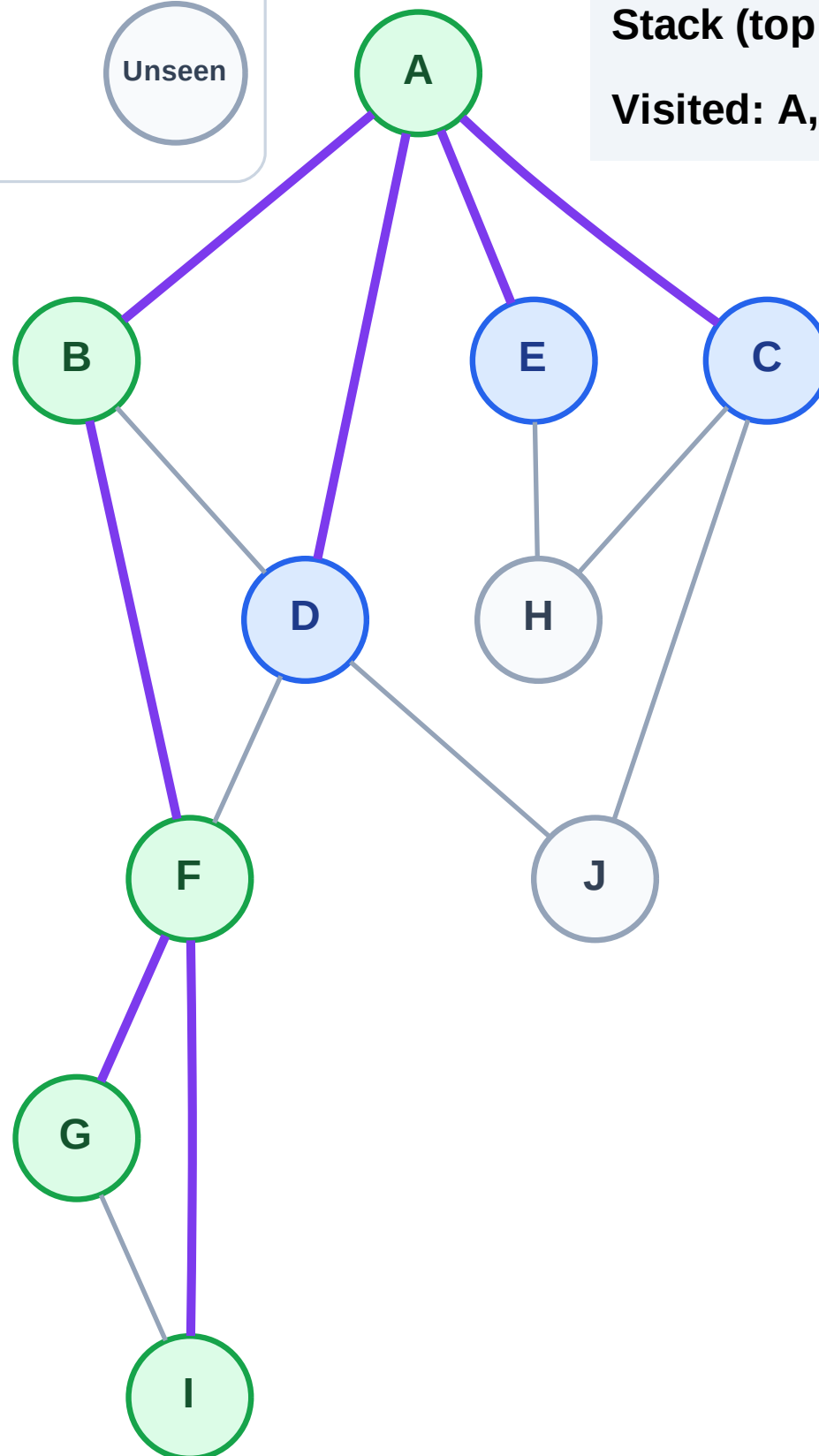
Step 11: No new neighbors from I

Legend



Stack (top at right): E | D | C

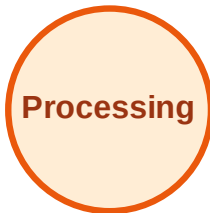
Visited: A, B, F, G, I



DFS Traversal

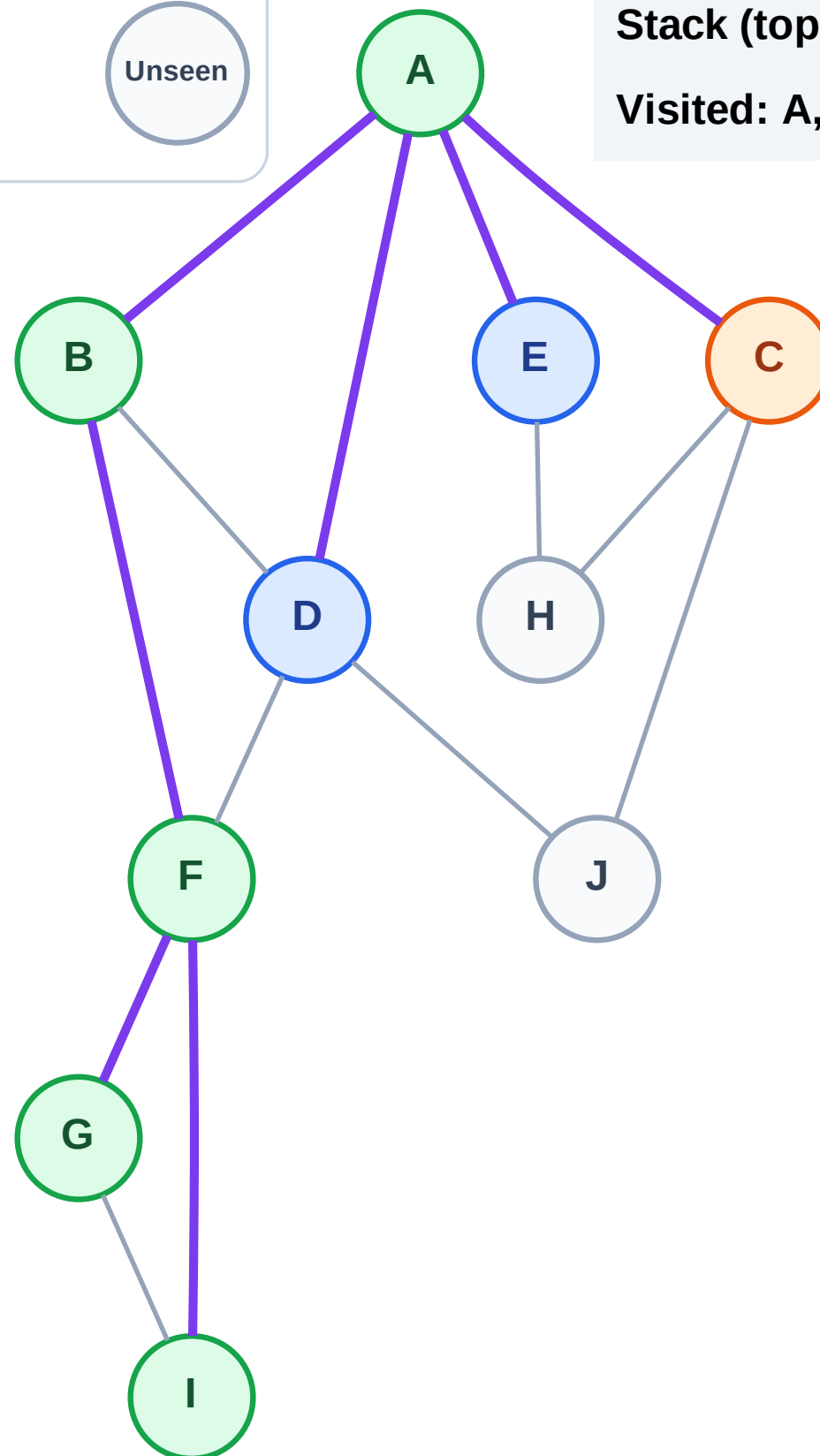
Step 12: Process node C

Legend



Stack (top at right): E | D

Visited: A, B, F, G, I



DFS Traversal

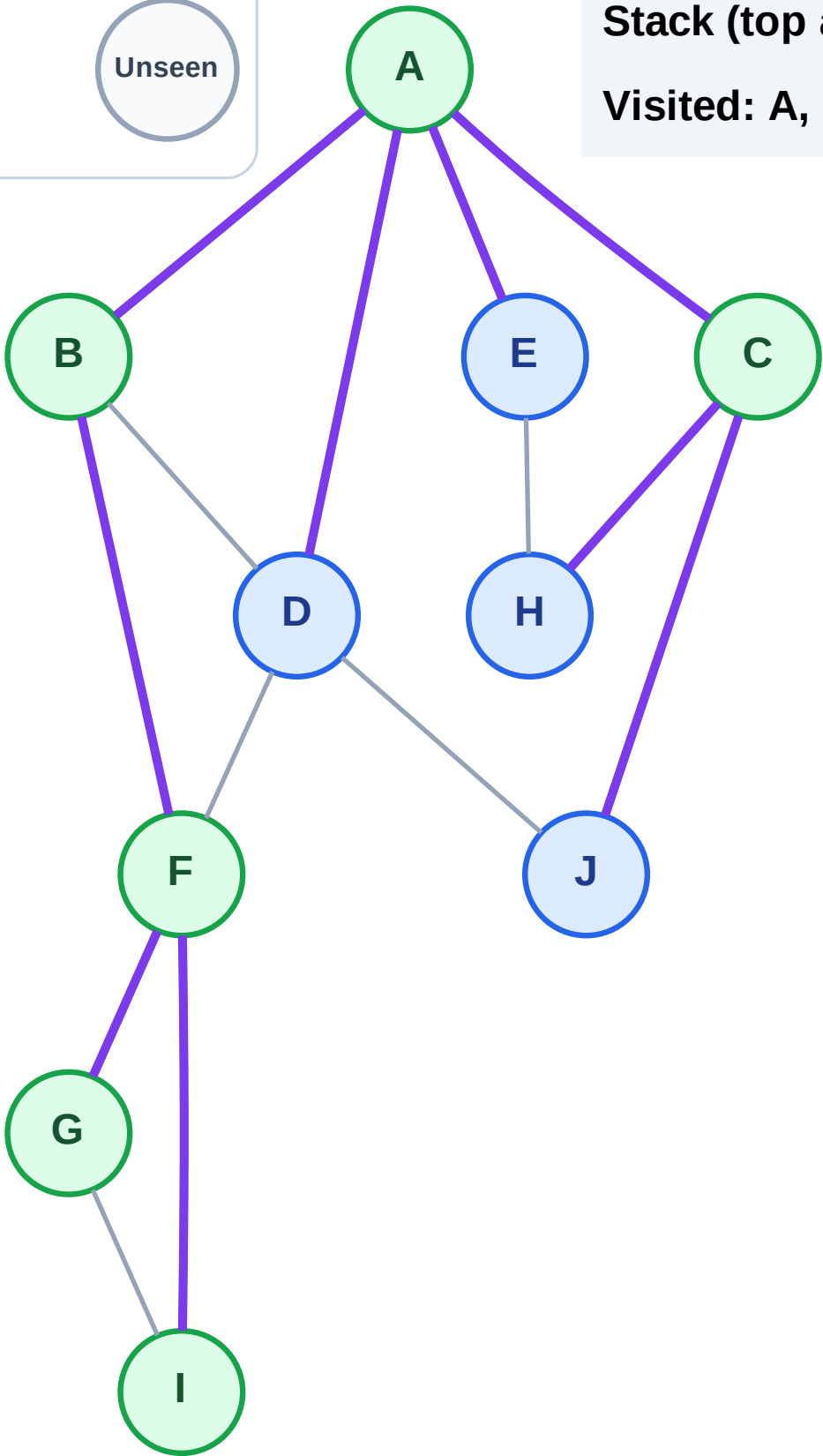
Step 13: Discover J, H from C

Legend



Stack (top at right): E | D | J | H

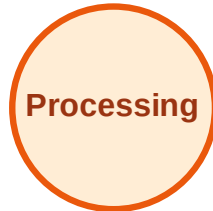
Visited: A, B, C, F, G, I



DFS Traversal

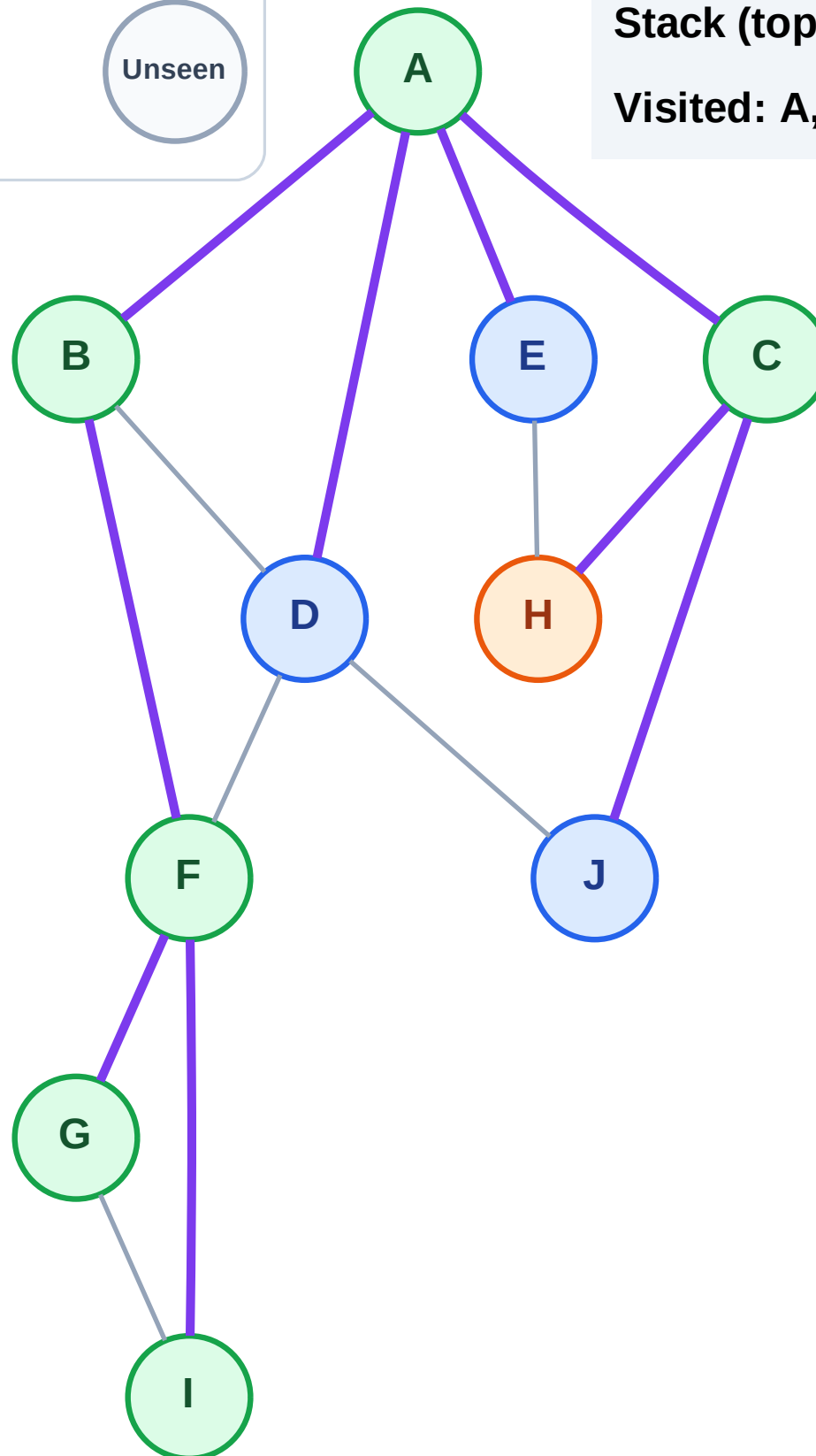
Step 14: Process node H

Legend



Stack (top at right): E | D | J

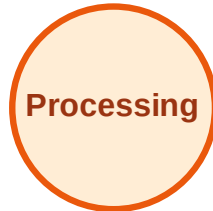
Visited: A, B, C, F, G, I



DFS Traversal

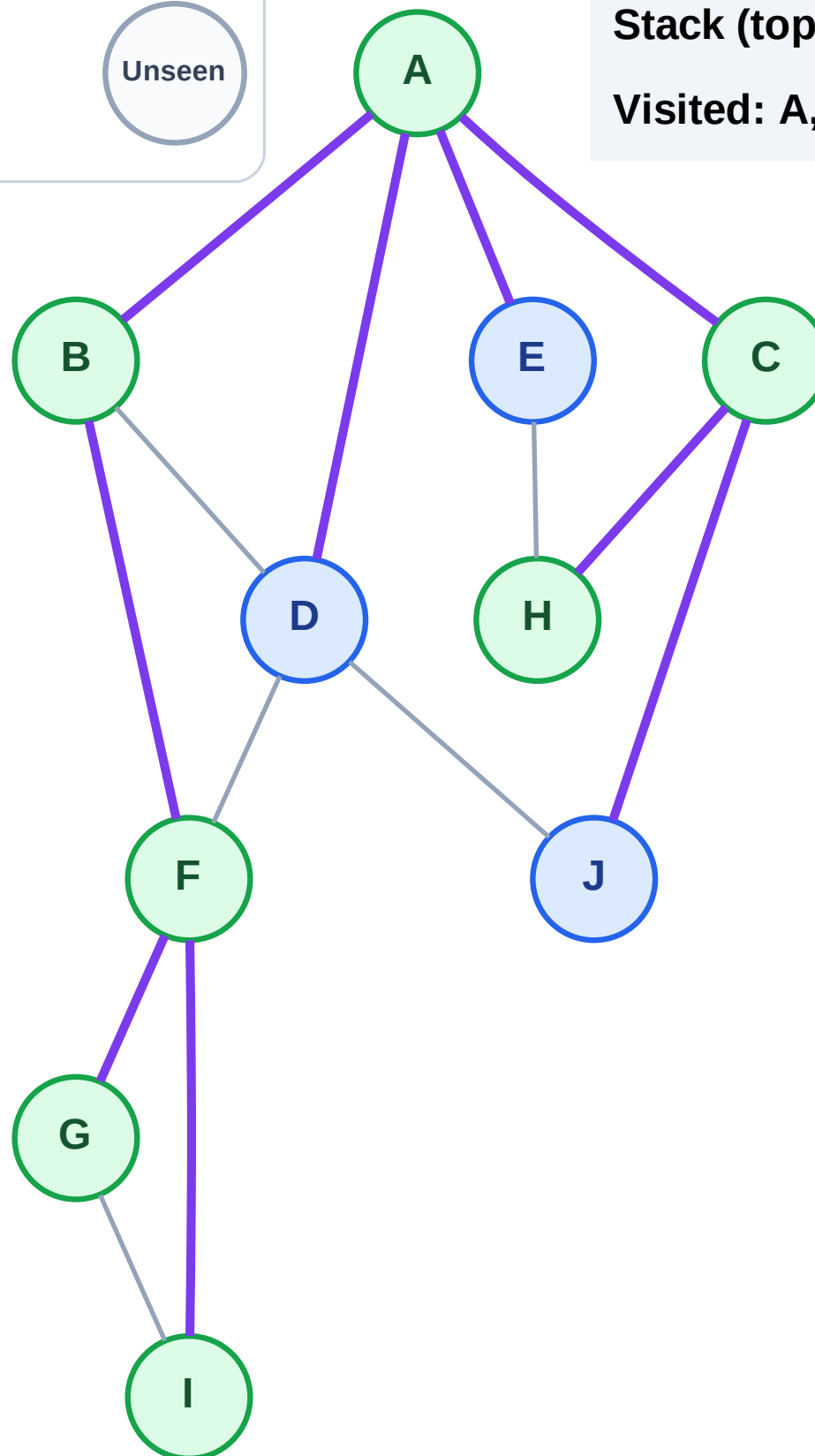
Step 15: No new neighbors from H

Legend



Stack (top at right): E | D | J

Visited: A, B, C, F, G, H, I



DFS Traversal

Step 16: Process node J

Legend

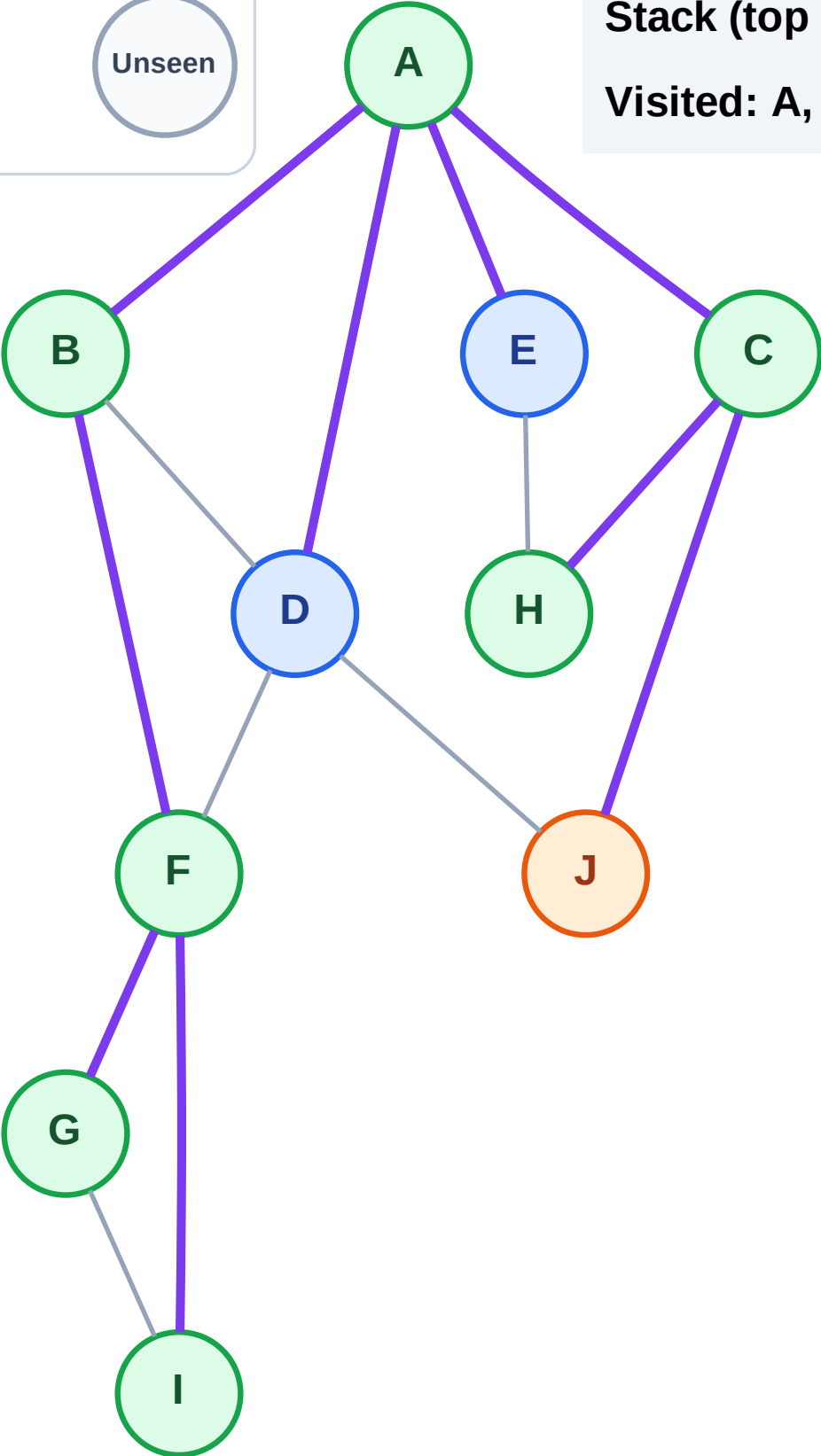
Complete

Processing

Discovered

Unseen

Stack (top at right): E | D
Visited: A, B, C, F, G, H, I



DFS Traversal

Step 17: No new neighbors from J

Legend

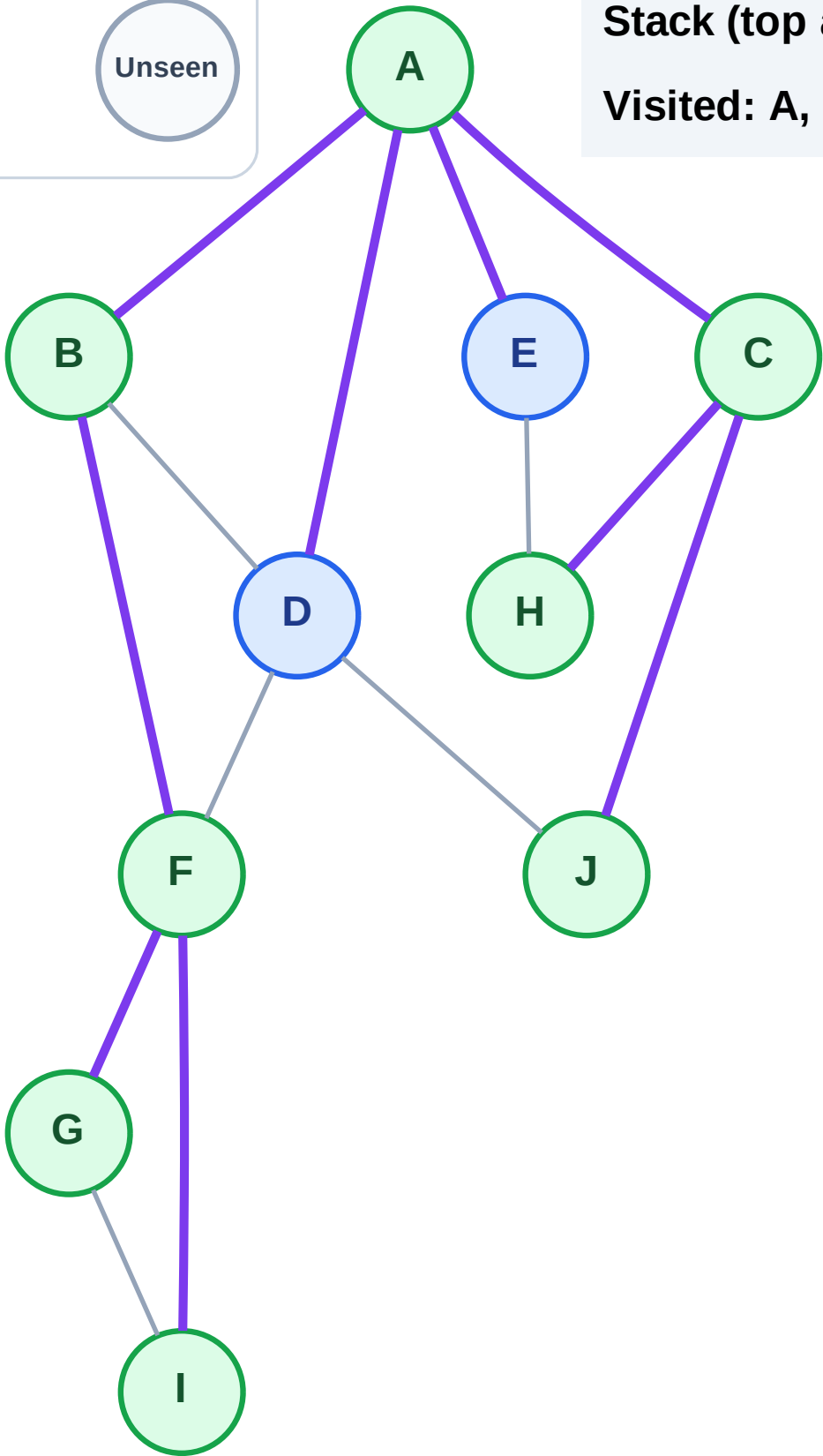
Complete

Processing

Discovered

Unseen

Stack (top at right): E | D
Visited: A, B, C, F, G, H, I, J



DFS Traversal

Step 18: Process node D

Legend

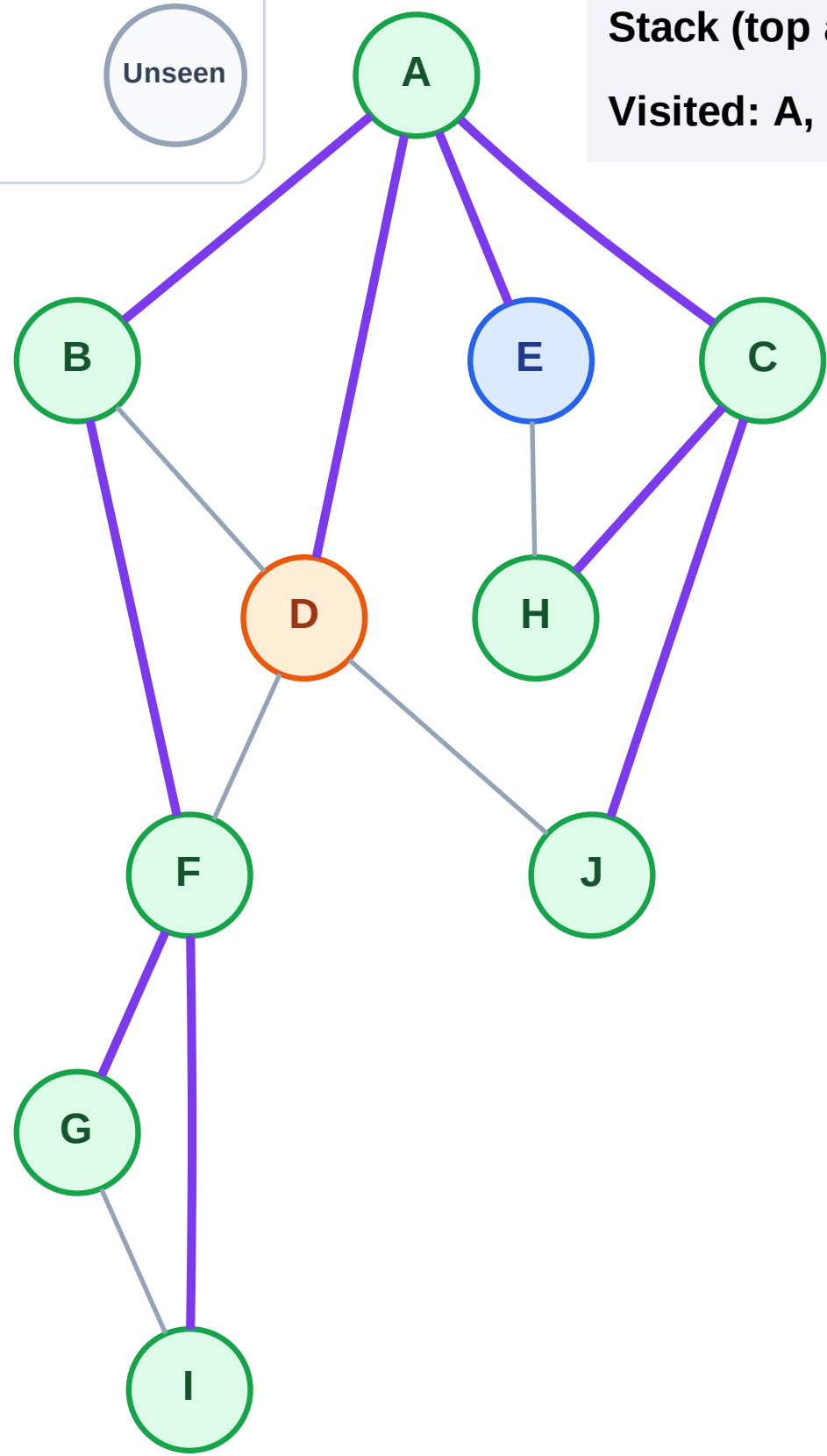
Complete

Processing

Discovered

Unseen

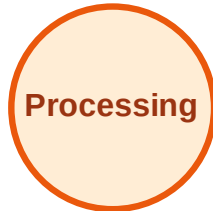
Stack (top at right): E
Visited: A, B, C, F, G, H, I, J



DFS Traversal

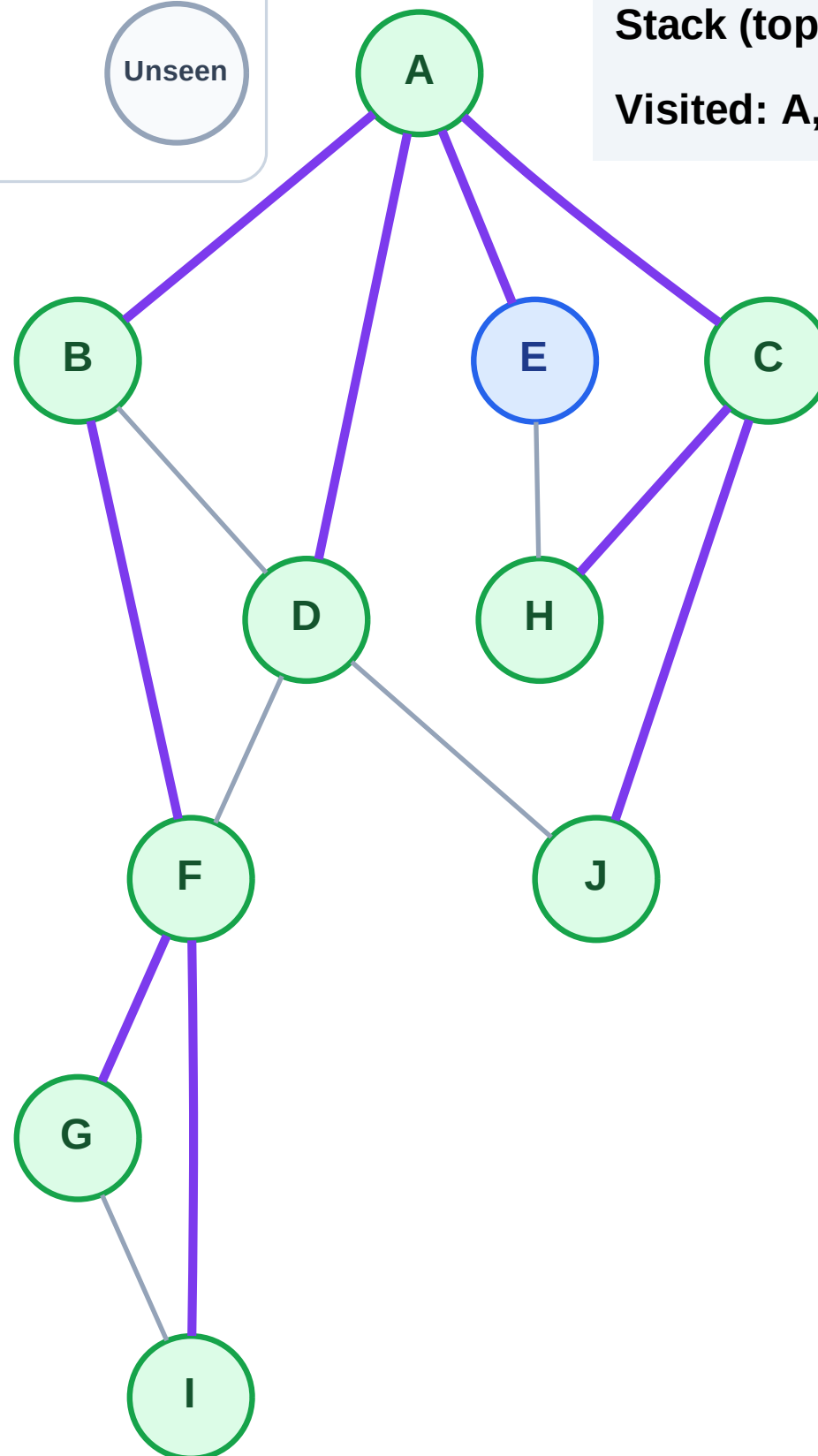
Step 19: No new neighbors from D

Legend



Stack (top at right): E

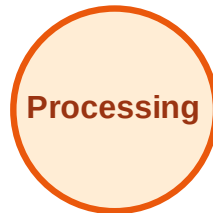
Visited: A, B, C, D, F, G, H, I, J



DFS Traversal

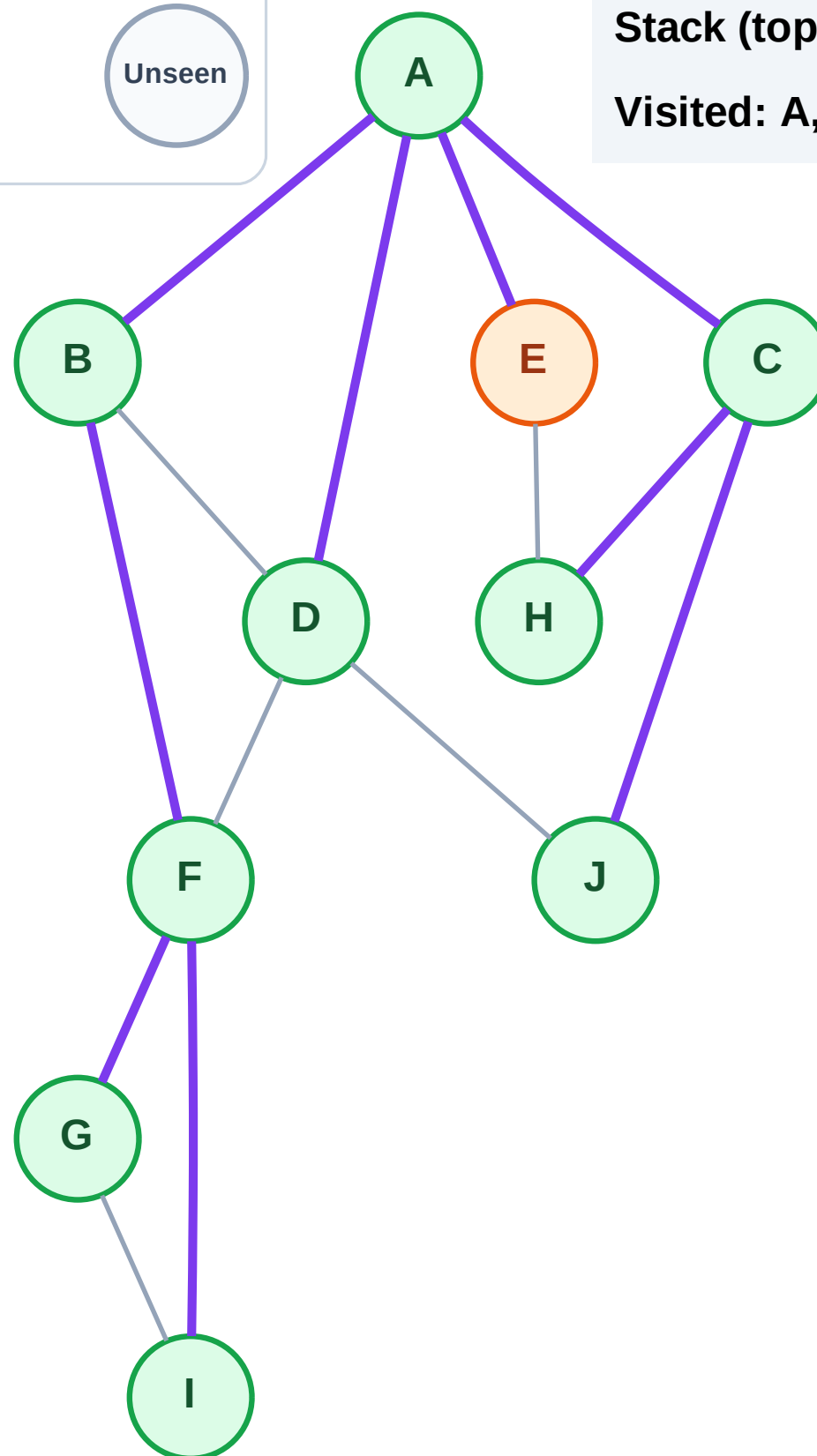
Step 20: Process node E

Legend



Stack (top at right): (empty)

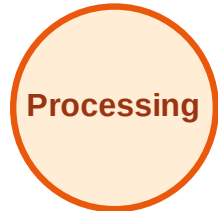
Visited: A, B, C, D, F, G, H, I, J



DFS Traversal

Step 21: No new neighbors from E

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

